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# Automating Electrical Grid Asset Inspection: From Current Challenges to Future Directions

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**ABSTRACT** Automated visual inspection systems are becoming increasingly important for improving the reliability, safety, and efficiency of electrical grid infrastructure. Despite significant advances in computer vision and artificial intelligence, inspection applications have not yet reached the performance of human experts. This review surveys current technologies, application domains, and methodological trends in automated grid asset inspection, while striving to uncover the root causes for the human-machine performance gap, advancing the state of the art through the analysis of common limitations and the formulation of future research directions. In addition, this work expands the scope of previous surveys beyond power lines to multiple electrical grid assets, acknowledging that most technologies and practices are shared across different application scenarios. The paper unique structure reflects the key dimensions for modeling an inspection system, ranging from inspected assets and tasks to model architectures and algorithms, including data acquisition platforms and sensors. This novel concept of defining a vision system through the intersection of independent domains promotes a system engineering perspective, which highlights often-overlooked issues that result from improper subsystem decomposition, manifesting as dataset limitations, inconsistent validation practices, and divergence between metrics optimization and operational testing. To address these issues and related problems, this work proposes a novel inspection system design, combining embedded vision for data validation, with multimodal datasets and models, and an active learning strategy for continuous model tuning. Overall, this paper provides a roadmap for developing inspection systems that combine accuracy, scalability, and cost-effectiveness, ultimately supporting maintenance and sustainability goals in electrical infrastructure management.

**INDEX TERMS** Active Learning, Asset Inspection, Computer Vision, Deep Learning, Electrical Grid, Embedded Vision, Multimodal, Photovoltaic, Power Line, Thermography.

## I. INTRODUCTION

ELECTRICAL Grid Asset Inspections have been drawing the attention of scientific and academic communities, with numerous research papers published on how to enhance the effectiveness of the inspection process by automating specific tasks. Confronted with various economic and political factors, the industrial sector is demanding effective solutions, while in the entrepreneurial sector, several innovative projects and pioneering companies are addressing the supply gap by committing to deliver efficient and viable inspection solutions.

This review examines common research trends and development strategies in Computer Vision (CV), recent advances in Deep Learning (DL), and the potential of knowledge transfer across domains, as well as challenges hindering the real-world adoption of proposed models. It also explores future directions for increasing the automation of electrical grid asset inspections. This study aims to align perspectives from both scientific and industrial sectors, fostering the development of new tools and technologies to benefit all stakeholders, including researchers, Distribution System Operators (DSOs), and society as a whole.

### A. SOCIO-ECONOMIC MOTIVATION

The EU DSO Entity, which represents Distribution System Operators at the European Union level, has been actively contributing to the implementation of EU energy policies [1]. Through its DSO Observatory, the organization monitors and analyzes emerging trends in the sector, offering critical insights into the challenges and opportunities faced by DSOs across the continent [2]. In response to increasing geopolitical instability, the European Commission launched the REPowerEU plan, designed to accelerate the shift towards renewable energy, reduce dependency on fossil fuel imports, and enhance energy security. Key policies under REPowerEU include the rapid expansion of renewable energy generation, the deployment of energy storage solutions, energy efficiency measures, as well as the promotion of energy-saving behaviors [3]. These ambitious targets, combined with emerging transformations, such as the electrification of key sectors such as transport and industry, place additional pressure not only on Transmission and Distribution (T&D) management systems but also on maintenance and inspection processes.

The global demand for electric power is projected to double by 2050, exceeding 50,000 TWh annually, with an expected 80% increase in wind and solar energy generation, according to the International Energy Agency [4]. This shift towards renewable and distributed energy resources necessitates extensive upgrades in T&D systems to facilitate the transition away from carbon-based fuels. In fact, over the next three decades, worldwide T&D systems are expected to exceed 160 million kilometers [4].

While developing efficient engineering processes for designing and constructing new infrastructure is vital to meet this growing demand, the management and maintenance of existing electrical generation and distribution systems are even more critical. The lack of reliability in these systems can have significant short-term impacts, such as frequent blackouts. In the European Union alone, over 50% of the grid has been in operation for more than 20 years, which is approximately half of its average lifespan [4].

Since operational failures due to wear and tear tend to increase at a growing rate over time, especially as assets approach the end of their expected service life [5], aging infrastructure poses an additional risk to grid stability and demands more frequent and targeted maintenance interventions. Among these aging components, insulators are particularly problematic, despite representing only $5 \sim 8\%$ of installation costs, they contribute to over 50% of maintenance expenses [6]. These facts highlight the urgent need for effective maintenance strategies to ensure the reliability of aging infrastructure in this sustainable energy transition.

In April 2022, the global solar power capacity reached 1 TW, which is equivalent[1] to 5 billion square meters ($5 \times 10^9$ m$^2$) of PhotoVoltaic (PV) surface area which also has to be subjected to periodic inspections. According to the National Renewable Energy Laboratory (NREL) technical report [7], the annual Operation and Maintenance (O&M) cost for utility-scale, single-axis tracking solar systems is approximately $16.42 per kilowatt (kW) of installed capacity, positioning the PV O&M sector as a billion-dollar industry.

Wind energy is also a significant contributor to this transition. As of 2023, the global installed capacity of wind power exceeds 900 GW, with an estimated 4 million wind turbine blades currently in operation, each measuring from 50 to 100 meters in length [8]. Regular inspection of these blades, which are constantly exposed to harsh weather conditions and mechanical fatigue, is essential to prevent potential structural failures, optimize overall performance, and ensure their long-term reliability.

Despite the slowdown in hydroelectric power plant construction, they still generate around 14% of the world's electrical power [9], and the installed capacity of over 1,400 GW in 2023 is expected to double by 2050 [10]. These facilities often feature large structures, particularly concrete dam walls, which can cover several thousand square meters and require regular inspection to ensure both safety and longevity, necessitating efficient solutions for data capture and processing.

With the rapid expansion of modern renewable systems and the continued dependency on traditional ones, manual image collection and analysis have become uneconomical and impractical. The current trend to address this bottleneck is to use remotely operated platforms equipped with various sensors to collect vast amounts of imagery data, which has led to an increasing demand for efficient data analysis.

According to Rosi's cost model [11], conducting inspection activities using traditional methods has a normalized cost of 266.11 €/km for *Electrical Power Lines* and 168.05 €/km for *Pipeline Systems*, respectively. When conducting inspection activities using aerial drones for data capture, there is a cost reduction of 66.0% and 45.8%, respectively. For *Photovoltaic* and *Wind Generation Systems*, the normalized costs are 295.79 €/MW and 85.29 €/MW, respectively, with cost reductions of 84.3% and 59.0% when using UAVs.

The growing reliance on electrical power across sectors highlights the need for enhanced grid reliability. Simultaneously, the integration of renewable energy sources and distributed generation adds operational complexity, increasing the demand for efficient O&M strategies. To address these challenges, the IEEE Power and Energy Society advocates for advanced asset management practices and automated inspection and maintenance methodologies, as outlined in the revised IEEE 1808-2024 guide [12].

Ensuring the safety and longevity of electrical assets through rigorous inspection schedules and proactive repairs is essential for supporting the goals of the REPowerEU plan [3]. These evolving challenges highlight the need for coordinated efforts at both the EU and DSO levels to standardize best practices in inspection, maintenance, and asset management. Such collaboration is crucial for empowering DSOs to address the dual challenges of expanding infrastructure while maintaining high reliability in an increasingly dynamic landscape of electrical energy production and distribution.

[1] Assuming an irradiance of 1000 W/m$_2$ and 20% conversion efficiency

### B. RELATED WORK

This section summarizes the document structure and focus of related review studies, highlighting some of their limitations that this work aims to address. To support one of the key contributions of this review—the identification of recurring patterns in resources and methodologies across different types of inspected assets—those review studies are organized by electrical system, ranging from power generation assets, including PV panels, Wind Turbine Blades (WTB), and Hydroelectric Dam Walls (HDW), to T&D power lines components.

In 2022, Høiaas *et al.* [13] surveyed imaging technologies for large-scale PV inspection, correlating PV module faults with power loss and thermographic patterns. The study emphasized the potential of luminescence methods (EL, PL, UVF) along with their respective operational constraints. Hijjawi *et al.* [14] extended this discussion, in 2023, by reviewing image-based and electrical measurement techniques for PV fault detection, with a comparative analysis of different data processing methods. Sepúlveda-Oviedo *et al.* [15] conducted a bibliometric review of PV fault diagnosis studies that included experimental results, clustering the works by Machine Learning (ML) paradigms—Symbolic, Bayesian, Analogical, Connectionist, and Evolutionary—based on the "schools of thought" framework followed by Domingos [16]. In 2024, Balachandran *et al.* [17] conducted a comprehensive review of inspection techniques used in PV fault diagnosis, describing various imaging technologies and general image preprocessing methods. Their study also outlined common fault causes, their visual and electrical signatures, and provided a brief overview of ML models and evaluation metrics.

The categorization used by Du *et al.* [18] in 2019 for WTBs damage detection techniques include: strain measurement, acoustic emission, ultrasound, vibration, thermography, and machine vision. In 2020, Marquez and Chacon [19] expanded on this by introducing radiographic, electromagnetic, shearography, image spectroscopy and data fusion. Dimitrova *et al.* (2022) [20] linked these techniques to Industry 4.0 [21], proposing digital twins for simulating extreme conditions and refining predictive maintenance strategies. This was further developed in 2024 by Memari *et al.* [22], who reviewed experimental studies monitoring structural responses and collect empirical data on fatigue performance under realistic loading conditions, for validating simulation models. While their extensive review was primarily focused on integrating UAV and DL technologies for WTB defect detection, particularly using visible and thermal imaging.

In 2023, Chen *et al.* [23] reviewed technologies for detecting underwater cracks on HDWs, covering methods such as visible light, sonar, acoustic, electromagnetic, and temperature-based techniques, along with the use of manned, remotely operated, and autonomous underwater vehicles. The study emphasized the limitations of traditional image processing under water and highlighted sonar's advantages in turbid conditions. Although DL techniques are well-suited to address these challenges, their deployment is hindered by the scarcity of annotated underwater datasets.

Early work by Matikainen *et al.* (2016) [24] reviewed remote sensing methods for monitoring Power Lines (PL) corridors, with emphasis on mapping and extracting components such as conductors and poles for vegetation encroachment assessment. Their taxonomy combined sensing platforms and technologies into a single dimension. Building on this, Nguyen *et al.* (2018) [25] provided a more structured review of inspection and navigation tasks. They defined three core components for inspection systems: the inspection method, the primary data source, and the analysis algorithm. Their tree-structured taxonomy covered inspection tasks and associated algorithms, but repeated terminology by combining platform and sensing technologies. Liu *et al.* (2020) [26] narrowed the scope further by focusing solely on visual inspection via UAV-collected RGB images, explicitly excluding other data types and platforms. Their study emphasized DL methods for component detection and fault diagnosis, offering summaries of popular techniques and public datasets. Sharma *et al.* (2023) [27] expanded the discussion by reviewing DL object detection methods for identifying PL components and related faults. Their taxonomy aligned with previous works and included brief overviews of inspection methods and data types. Although they compiled a comprehensive list of model improvements, organizing them into broader categories could enhance clarity. Their emphasis on dataset importance is valuable, though a deeper examination of dataset quality and generalization challenges would add further insight. Most recently, Anderson and Kiamilev (2024) [28] provided a concise overview of vision-based inspection, highlighting opportunities enabled by emerging technologies. They classified data acquisition into manual and automated methods and categorized data processing into component detection, defect detection, and condition assessment.

A common limitation across related reviews is the redundancy introduced by treating combinations of platforms and sensing technologies as distinct categories. For instance, UAV-based and airborne LiDAR are often considered separately, despite relying on the same underlying sensor. DL has emerged as the dominant processing approach, partly driven by the cost-efficiency of UAV-based data collection. While processing methods are not inherently tied to specific acquisition procedures, they are critically linked through data annotations—their quality and structure play a central role in enabling effective analysis, yet remain underexplored in the literature. Moreover, limiting the review of data processing methods used within the electrical grid domain overlooks transferable advances from broader CV research, risking the omission of state-of-the-art methods that could enhance grid-related performance. Another frequent shortcoming is the limited consideration of real-world applicability. Related works often fail to address integration constraints, or operational viability that ultimately determine their practical value in industrial settings. By addressing these issues, this review aims to provide a more structured and application-oriented perspective that better aligns with the practical demands of the electrical grid industry.

### C. REVIEW METHODOLOGY

This review employs a systematic search strategy combined with a targeted critical evaluation of studies. While searches were conducted across multiple databases, final study selection prioritized methodological rigor, clarity of experimental validation, and inspection task diversity, aligning the review with a focused critical approach.

Searches were conducted across multiple databases (*Web of Science*, *ScienceDirect*, *IEEE Xplore*) using an advanced Boolean query. Publications from 2015 to 2024 were considered, reflecting the period in which DL methods became practically feasible and influential. The increasing number of publications over this period is illustrated inFig. 1.

To capture all relevant variations of search terms, results for each electrical grid asset were obtained from *Web of Science*, with the *"exact search"* option disabled to allow for variations such as plurals and other suffixes. Boolean conditions were applied to simultaneously (AND) include the topics of interest, along with alternatively (OR) related synonyms, as formulated in Equation 1, where logical operations (AND, OR) over a series of terms are represented by $\prod$ and $\sum$, respectively. The index $i$ denotes each keyword group, while $j$ corresponds to individual keywords within a group.

$$\text{Search Query} = \prod_{i=1}^{n} \left( \sum_{j=1}^{m_i} \text{keyword}_{i,j} \right) \tag{1}$$

Table 1 contains the set of keyword terms used to obtain the *Power Lines* data series counting results. Each column $i$ represents a logical AND input, and each row $j$ represents a logical OR input. The other series quantified in Fig. 1 were obtained by modifying only the fifth column with respective synonyms for *Voltage Insulators*, *Substations*, *Photovoltaics*, *Wind Turbines*, and *Hydroelectric Dams*.

**TABLE 1. Keywords used in *Power Lines* search query**

| | | | | |
|---|---|---|---|---|
| Visual<br>Vision<br>Optical<br>Image | Inspection<br>Inspect<br>Detect<br>Identify | Defect<br>Fault<br>Anomaly | Deep Learning<br>Machine Learning<br>Artificial Intelligence<br>Computer Vision<br>Image Processing | Power Line<br>Transmission Line<br>Electric Line |

The identification of additional studies was further supplemented by following references and citation networks, ensuring broader coverage of relevant literature. The resulting sample represents only a subset of the overall population, acknowledging that some relevant works may not have been captured by the database search or supplementary tracing.

Despite the exponential growth in electrical grid asset inspection research, as shown in Fig.1, it is important to note that global education expansion has contributed to rising publication rates across various fields [29]. On the other hand, it can be argued that the growth in automating visual inspection of defects in electrical grid assets has outpaced the general trend observed in other areas of study. However, if the growth or research in this field was normalized, it would appear less pronounced than in the raw trend presented in Fig. 1.

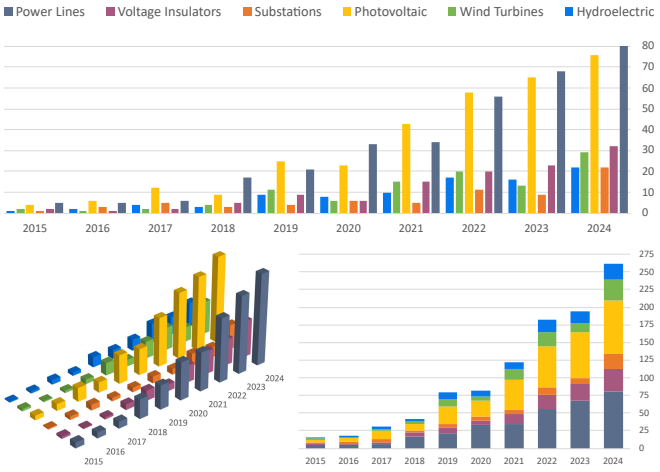



**FIGURE 1. Different illustrations of the count of published articles on electrical grid asset visual inspection over the past decade.**

To ensure transparency and reproducibility in the study selection process, the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 framework was followed. The flow diagram in Fig. 2 summarizes the identification, screening, eligibility assessment, and final inclusion of studies, datasets, and models. Duplicates and out-of-range records were removed, titles and abstracts were screened for relevance, and full texts were assessed based on the use of imagery data, application of deep learning, and relevance to electrical grid asset inspection.

For the focused critical review, candidate studies were ranked by citation count and journal quality and selected based on methodological rigor, task diversity, and clarity of experimental validation. This approach prioritizes quality and relevance over quantity, acknowledging that the selected sample represents a subset of the broader literature.

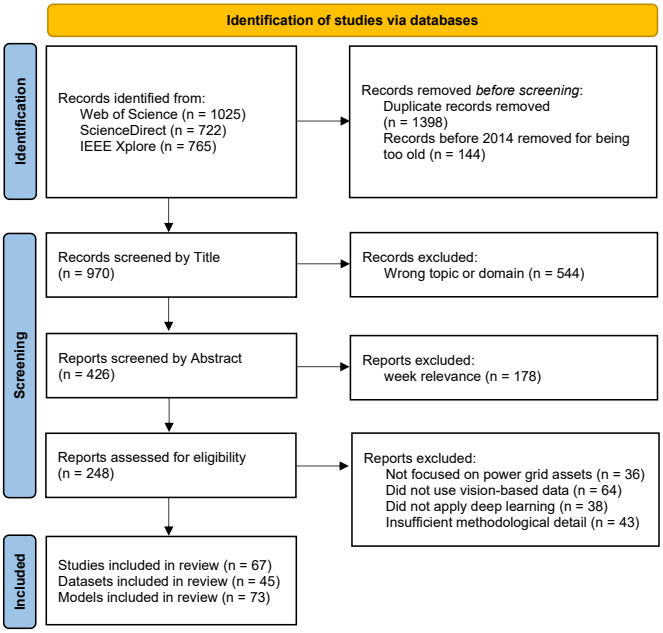



**FIGURE 2. PRISMA 2020 flow diagram of the literature selection process, showing the number of records identified, screened, assessed for eligibility, and included in the review.**

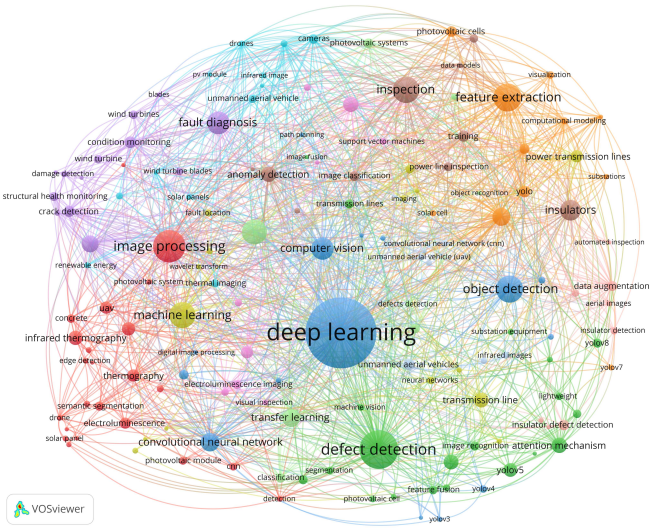



**FIGURE 3. Network visualization of co-occurring keywords in the computer vision and power line inspection literature, generated with VOSviewer. Node size represents keyword frequency, and link thickness indicates co-occurrence strength.**

To complement the systematic search and gain insight into the research landscape of vision-based grid asset inspection, a bibliometric analysis was conducted to map keyword relationships and identify dominant trends. Figure 3 illustrates the co-occurrence of keywords, highlighting the main research clusters, emerging topics, and less developed areas.

Deep learning dominates as the methodological core, strongly linked with object detection, computer vision, and defect detection. Image processing, machine learning, and thermography form a secondary cluster representing classical approaches. Application-oriented clusters emphasize inspection, feature extraction, insulators, and transmission lines, while fault diagnosis, condition monitoring, and renewables (wind, solar) reflect diversification of targets.

Dense ties between deep learning, object detection, and defect detection indicate a mature research axis, while links with image processing show an ongoing shift from classical to DL methods. Fault diagnosis connects mainly to renewables, highlighting growth beyond transmission lines. The clustering of YOLO variants (v3–v8) around detection tasks signals methodological consolidation around real-time DL models.

Trends include the centralization of research around deep learning detection frameworks and increasing attention to renewable energy assets. Gaps are visible in the weak coupling of fault diagnosis with operational decision-making, and in the limited cross-sensor integration (e.g., LiDAR, multispectral), pointing to opportunities for multimodal approaches.

### D. MAIN CONTRIBUTIONS

This review builds on previous works by covering distinct but related electrical grid assets and summarizing information both broadly, across sectors and applications, and in depth, through technical details and case studies. The resulting survey provides a refined perspective, consolidating existing knowledge and highlighting emerging areas of interest.

The main contributions are summarized as follows:

- **Multi-asset Coverage:** Expands the review beyond power lines to multiple electrical grid assets, acknowledging that most technologies and challenges are shared across scenarios.
- **Perspectives Integration:** Combines academic research with condition monitoring and maintenance management perspectives, analyzing both technological advancements and industrial case studies to achieve unique insights.
- **Framework Conceptualization:** Introduces a multi-dimensional axis illustrating the methodology used to describe the combined domains for modeling an inspection system, structuring the analysis systematically while reducing redundancy in platform-sensor categorization.
- **Analytical Digestion:** Consolidates over 250 references, with complex relationships and processes summarized through 12 diagrams, alongside an in-depth synthesis of 67 core works, complemented by comparative analyses of 45 datasets and 73 models, grouped by inspected asset across 9 tables.
- **Key Challenges Identification:** Highlights bottlenecks such as dataset limitations, annotation quality, inconsistent validation practices, and gaps between model optimization metrics and real-world operational testing. These insights outline promising directions and guide future research.
- **Integrated Proposal:** Presents an integrated development methodology combining embedded vision for dataset validation, multimodal domain transfer, and active learning. This approach enhances operational applicability, enabling a scalable and deployable inspection system.

Prior reviews often suffer from key limitations. From the identified patterns in related works (Sec. I-B), a suboptimal organizational tree-structure methodology was identified, which promotes redundancy in the exposure of information. To address this issue, a different approach was followed, encompassing the relationships between the considered variables in a multi-dimensional space.

In this study, the dimensions represent the key factors considered in defining the computer vision model. Each dimension is associated with a domain, which defines the valid set of entities for that dimension. The four main dimensions considered are **Assets**, **Means**, **Sensors**, and **Algorithms**, with their respective domain definitions described as follows:

- **Assets** - Physical structures, devices, equipment, and components that are the objects of inspection. This includes power line conductors, voltage insulators, substation switchgear, photovoltaic modules, wind turbine blades, and hydroelectric dam walls.
- **Means** - Platforms and other means of mobility that constrain the position (distance and angle) between the sensorial technology employed and the inspected assets, including manned and unmanned vehicles, terrestrial and aerial, as well as satellites and foot patrol.
- **Sensors** - Passive and active optical instruments that measure the assets' irradiation energy, producing a signal with spatial relations, such as RGB-D cameras, thermographic, ultraviolet, and LiDAR sensors.

- **Algorithms** - Computational methods designed to interpret and analyze visual information from the real world based on deep learning, including advanced artificial neural network architectures and learning paradigms.

To better understand the conceptual relationships between each dimension, complementary sub-dimensions are introduced, such as the task dimension as the combination of assets and means, and the data dimension as the combination of task and sensors. The categorical dimensions considered, namely, inspection **Task**, acquired **Data**, and deep vision **Model**, are graphically represented in Fig. 4 in two-dimensional input space, and in multi-dimensional input space in Fig. 5.

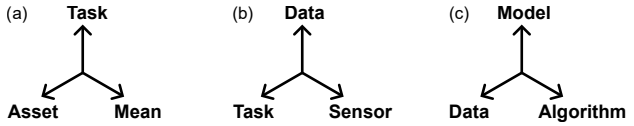



**FIGURE 4. Considered dimensions in two-dimensional input space (a) Inspection Task (b) Acquired Data (c) DV Model.**

For instance, the model developed depends on both the algorithm employed and the dataset selected for training. In turn, the data depends not only on the sensor used but also on the intended task, which is reflected, among other factors, in the annotations. Finally, the inspection task is determined by the possible defects of the asset and the appropriate means employed for conducting the inspection effectively.

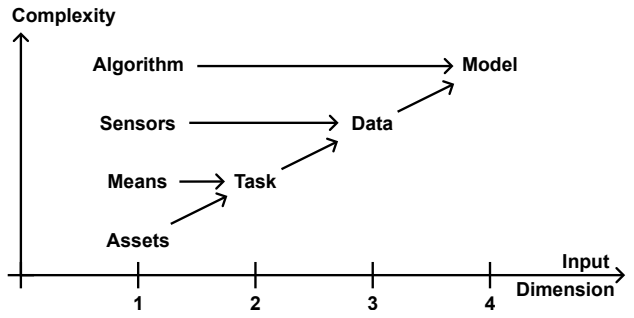



**FIGURE 5. Relationships and classification of considered system dimensions. Input Dimension axis ranks each domain by the number of its primary dependencies, while the intrinsic combination of these dependencies increases the dimension Complexity on the vertical axis.**

The introduction of this unique convergence of categorical dimensions that define the solution for automating electrical grid asset inspections as a four-dimensional input problem was driven by a Systems Engineering (SE) [30] approach and an Enterprise Architecture (EA) [31] perspective, which focus on the effective breakdown of complex problems into subsystems, while aligning the solution with broader organizational goals, thereby ensuring a comprehensive and efficient solution design. This methodology eliminates redundant information to maintain focus and clarity by identifying and analyzing patterns across various development strategies presented in the literature. It is important to note that this multi-dimensional tensor is not fully populated, and this sparsity represents an acceptable intentional effort to eliminate the redundancy inherent in the typical tree structure adopted by other authors.

### E. DOCUMENT STRUCTURE

This review starts, in Section I, by introducing the socio-economic motivation for studying the development of efficient inspection systems, discussing related works, describing the review methodology, outlining the main contributions, and synthesizing the document structure, illustrated in Fig. 6.

Section II defines the visual inspection process, characterizing the electrical systems considered and the inspected assets. It also defines different levels of inspection process automation and discusses the factors that influence its profitability.

Section III addresses the data acquisition procedure, summarizing the technologies and sensors used, comparing the means and platforms available, and aggregating the operations conducted by the defects scale and inspection distance.

Section IV focuses on data processing, providing an overview of the historical evolution in deep vision, highlighting the main stages of learning model development, and introducing emerging paradigms and their potential.

Section V explores the application scenarios of deep vision inspection systems from a practical perspective, recognizing popular strategies while avoiding detailed discussions of algorithms to prevent information repetition. It begins with power generation systems, such as photovoltaic modules, wind turbine blades, and hydroelectric dams, and concludes with transport and distribution systems, ranging from power line mapping and structure identification, HV insulators and other components, to substations and transformers.

Section VI highlights the challenges and limitations faced by practitioners and researchers, while section VII proposes a system's level framework for automate asset inspection, providing directions for future research. Finally, section VIII provides a conclusion by briefly summarizing the starting point, the ongoing development, and the goals to be attained.

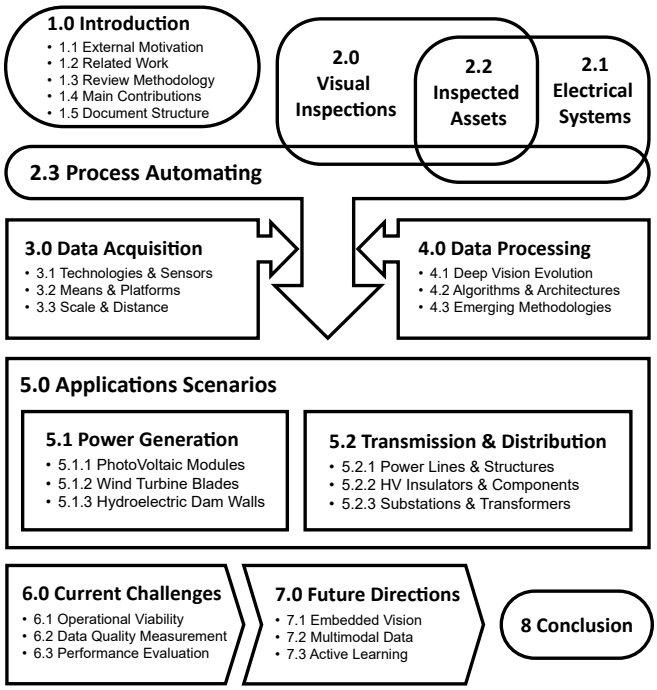



**FIGURE 6. Overview of the document structure conceptualization.**

## II. VISUAL INSPECTIONS

Often considered the most basic and direct form of Non-Destructive Testing (NDT) methods, visual inspection is used to evaluate the condition of an asset as part of the overall maintenance process [32]. Traditionally, this inspection method comprehends the analysis of naked eye visual spectrum information, without special equipment but requiring special training. The observer's visual perception is used to identify visible defects, anomalies, wear and tear, or other forms of damage, such as cracks, corrosion, leaks, or irregularities in a product or structure that could potentially impact its functionality, safety, or quality. The primary goal of an inspection is to identify prematurely minor issues and their causes in order to prevent more significant and costly damage or consequences.

While visual inspections are valuable for detecting surface-level issues, they may not uncover defects hidden beneath the surface. As a result, they are often complemented by other NDT techniques, including *Radiography* or *Thermography*, which expand the inspector's observability of the electromagnetic spectrum, beyond the human visible range; *Magnetic Testing* or *Eddy Current Testing*, which is employed to identify structural imperfections such as cracks and voids by examining disruptions in the flow of the magnetic or electric field; *Acoustic* or *Ultrasonic Testing*, which involves comparing the material's sound propagation with a reference or baseline signal obtained from a defect-free asset [33].

Advancements in technology have eliminated the need for in-person visual inspections, expanding the scope of current practices. Digital cameras make it possible to collect large amounts of high-detail data locally, which can then be efficiently analyzed remotely. This is particularly useful in harsh environments such as mines, storage tanks, or industrial boilers. The use of drones and other robots facilitated the access and collection of data on remote locations such as wind turbine plants, offshore platforms, or submersible structures, as well as confined spaces like pipelines, ducts, or pits.

Modern visual inspections integrate data analysis technologies, such as Computer Vision (CV) models. CV is a scientific field focused on enabling artificial vision. Its origins trace back to the invention of the photographic camera in the 19$^{th}$ century, followed by the television. The term "computer vision" emerged in the 1960s to describe mimicking the human visual system, with studies on image processing algorithms surging in the 1970s and optical character recognition (OCR) developed in the 1980s [34]. Since then, CV has evolved at an exponential rate, impacting fields from military and medical sectors to industry and mobility. CV systems aim to overcome human vision's limitations, expanding its capabilities. One goal is to translate information between domains by processing signals outside the visual electromagnetic spectrum for human interpretation, examples include SoNAR, LiDAR, MRI scans, meteorological satellites, and radio telescopes. Another objective is to emulate visual processing, by designing algorithmic functions, in order to automate decisions, overcoming human limitations like fatigue and subjectivity [35].

In an automated grid asset inspection system, a CV model is defined by the algorithm used to process the data. The data are determined by the sensor used to acquire the signal and the annotation guidelines implicit in the task. The inspection task is primarily concerned with the assets to be inspected, as well as the means and platforms employed. The relationships between these system dimensions are graphically represented in Fig. 5, with examples for each domain provided in Fig. 7.

Capturing these interdependencies is essential for engineering an effective inspection system. Isolating the model from other subsystems requires careful interface design, as emergent properties [36] affect the engineering of complex systems. Most recent publications have focused on a model-centric approach, improving neural network architectures or learning algorithms [37]. However, the importance of data has often been underestimated, leading to advancements that do not align with industry expectations.

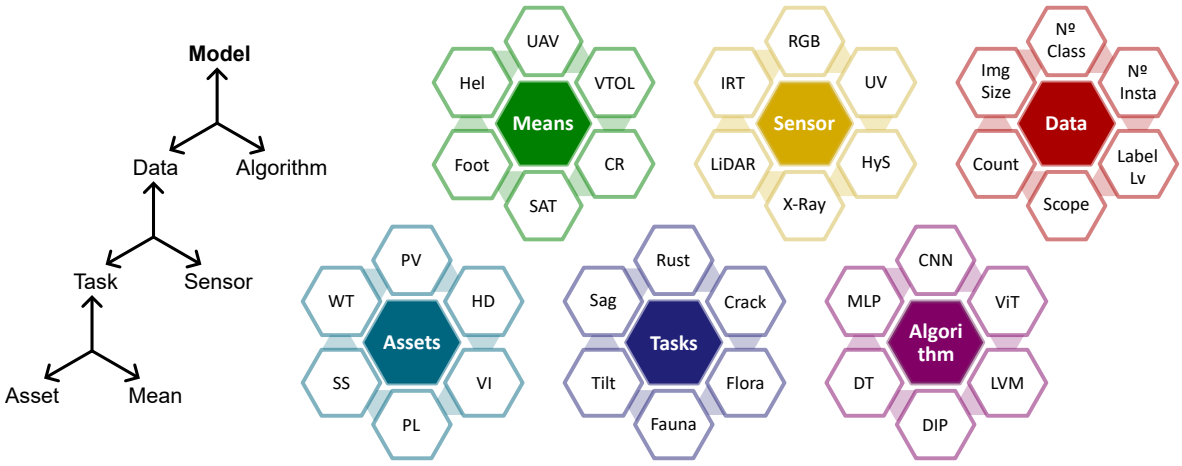



**FIGURE 7.** Overview of the existing dimensions and corresponding domain samples relevant to inspection system modeling. Assets domain includes: Photovoltaics, Wind Turbines, Hydroelectric Dams, Power Lines, Voltage Insulators, and Substations. Means domain includes: UAVs, Helicopters, VTOL Aircrafts, Foot Patrols, Satellites, and Climbing Robots. Tasks domain includes detection of: Rust, Cracks, Vegetation, Bird Nests, Tilting of structures, and Sagging of power cables. Sensors domain includes: Visible Image, Infrared Thermography, Ultraviolet Imaging, LiDAR, X-Ray, and Hyperspectral. Data characteristics domain includes: Image Size, Number of images, Classes and instances, Level of label detail, and data Scope diversity. Algorithms domain includes: Digital Image Processing, Decision Trees, Multi-Layer Perceptrons, Convolutional Neural Networks, Vision Transformers, and Large Vision Models.

### A. ELECTRICAL SYSTEMS

Electrical systems, ranging from power generation to T&D, rely on a variety of components to ensure the safe and efficient production and transmission of power. Among these components, PV panels are vital contributors to meeting the growing global demand for renewable energy [4]. In addition, high-voltage (HV) insulators—responsible for over 50% of T&D maintenance costs [6]—play a crucial role in preventing electrical faults by maintaining the necessary electrical separation between conductive supporting structures and active elements.

Besides describing the electrical principles of PV panels and HV insulators, this section summarizes their materials, performance characteristics, and common failure mechanisms. Understanding these concepts is crucial for optimizing inspection and maintenance strategies through the development of reliable and effective CV systems. Although wind turbines (WTs) and hydroelectric dams (HDs) are included in this study as power generation systems, the respective surface-exposed assets with large and repetitive structures—namely WT blades and HD walls—are passive elements. Thus, they differ fundamentally from PV modules, which are not only visually accessible but also functionally active.

#### 1) Photovoltaic Systems

PV power stations or simply PV plants, comprise a set of PV arrays, each consisting of multiple PV panels, made up of several PV modules, which are assemblies of interconnected PV cells. Crystalline Silicon (c-Si) modules, made from silicon wafers, are efficient and durable but costly and resource-intensive to produce. Cadmium Telluride (CdTe) thin-film modules are lighter, more flexible, and cheaper to produce, with lower material and energy demands.

A PV system failure occurs when the photovoltaic components cannot function optimally or sustain their intended performance levels. These failures result from both external and internal causes. External factors include issues during transport, poor installation, and exposure to extreme conditions, such as excessive dust, soiling, or vegetation interference. Internal failures stem from structural issues, such as cracks in PV cells or faults in cell connections [38].

Photons striking the PV cell excite electrons, allowing them to jump from the valence to the conduction band and form electron-hole pairs. These pairs separate at the N–P junction, where the electric field drives electrons toward the N-type side. This movement reduces the depletion zone and generates an electric potential across the cell's electrodes [39].

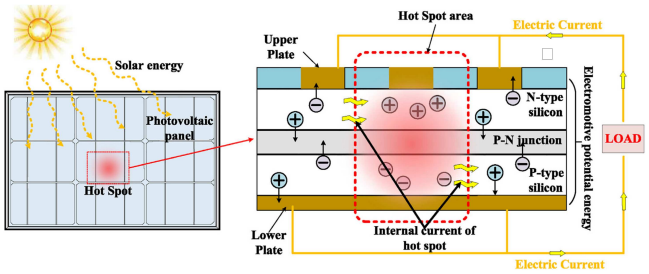



**FIGURE 8.** Overview of hotspot formation in PV cells, adapted from [40].

Hot spots in PV panels occur when normal current flow within silicon semiconductor cells is disrupted, often due to blocked photoelectric effect areas, as shown in Fig. 8. These blockages may stem from internal damage or pronounced irradiance contrasts caused by objects shading the cells. This reverses the current flow, causing shaded cells to act as resistors in the panel circuit, consuming energy instead of producing it, resulting in localized temperature increases.

Prolonged exposure to hot spots can damage PV cells, leading to burnout or degradation, and in severe cases, cell failure, which disrupts the entire panel's functionality. Hot spots reduce efficiency and pose fire risks. Bypass diodes in solar panels mitigate partial shading effects by redirecting current around shaded areas, preventing excessive heat buildup.

The adverse effect of vegetation-induced shading on a PV module performance was analyzed by Starzy ski *et al.* [41]. In this simulation, the slightly shaded module experienced a disproportionate power drop of over 45%, decreasing from 365 W to 199 W. In addition, this reduction in output power from a single module propagates throughout the entire string of series-connected modules, leading to significant overall losses. The study highlights the cumulative impact of shading, noting that even minimal shading can reduce PV panel efficiency by up to 10%. However, modules equipped with Half-Cut technology [42] tend to experience a lower impact from shading compared to standard modules [41].

#### 2) Power Line Insulators

PL Insulators are essential components for the reliable suspension of transmission line conductors, which must withstand mechanical, electrical, and environmental stresses.

Insulators can be classified based on their material composition, format, and function [43]. One of the most commonly used insulator type is the *Cap & Pin* disc, with a form illustrated in Fig. 9. Each unit is constructed of a ceramic or glass disc with a metal cap and pin cemented to opposite sides, and connected to the next, by sliding the steel pin into the steel cap cavity, forming an insulator string that can typically support up to 120 kN ($\approx$ 12 tonnes) of mechanical tension.

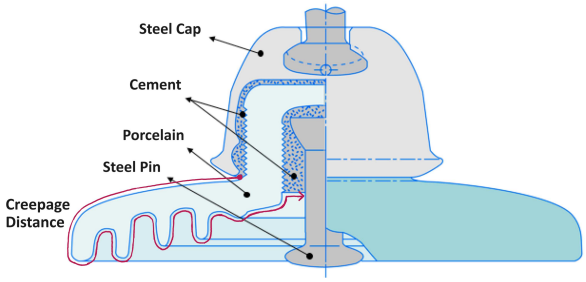



**FIGURE 9.** Overview of a typical *Cap & Pin* disc insulator shape and construction details, adapted from [44]. The creepage distance (the shortest path along a surface between metallic parts) is often extended by designing protrusions on the underside of the disc.

When exposed to the environment, the hydrophobic surface of ceramic insulators becomes contaminated by coastal pollution, including sand and salt carried by sea winds, or other deposits from industrial sources. These contaminants increase

surface conductance by forming a conducting electrolyte, especially in moist weather conditions. While light rain and fog exacerbate the issue, heavy rain helps in removing some debris. The choice of the insulators specific creepage length depends on the severity of pollution at the site [45].

Air ionization can occur if the electrical field strength exceeds 3 kV/mm at atmospheric pressure. The voltage required to cause flashover in a dry clean single insulator with a 280 mm creepage distance is 72 kV. However, under medium pollution conditions, it is estimated that its performance decreases tenfold [46]. This significant reduction is due to the presence of a conductive layer on the insulator's surface. Leakage current flowing over the insulator surface, along with its heating effect, causes the formation of dry bands or stains, as shown in Fig. 10, further reducing shell's hydrophobicity.



**FIGURE 10.** Example of insulator flashover damage, adapted from [47].

Ceramic and polymer insulators are engineered to withstand higher puncture voltages compared to flashover voltages, ensuring that they experience flashover before puncturing. In contrast, glass insulators are designed in such a way that an overvoltage results in a puncture arc through the glass, rather than a flashover. To facilitate the identification of defective units, glass insulators undergo heat treatment, which causes the glass to shatter upon puncture, making the absence of the damaged disc more apparent. This approach is particularly beneficial, as markings on glass could otherwise be difficult to detect. Despite this, the mechanical strength of the unit remains unaffected, ensuring that the insulator string remains intact even in the event of failure.

### B. INSPECTED ASSETS

Electrical assets in power generation systems, such as PV modules, wind turbine blades, and hydroelectric dam walls, are typically large with extensive surface areas, requiring data acquisition from multiple positions while maintaining fixed distances and angles relative to the asset. In contrast, T&D assets, like power line conductors, high-voltage insulators, substations, and transformers, exhibit greater variability in size, shape, and positioning. While many of these assets have smaller surface areas, their irregular, three-dimensional forms require varying inspection angles and distances, necessitating data capture from multi perspectives for thorough coverage.

Industry standards are essential in ensuring effective and accurate inspections for both power generation and T&D assets, providing structured frameworks for data collection, analysis, and assessment, ensuring consistency and reliability.

The IEEE 1808 standard [12] provides guidelines for collecting and managing inspection data, ensuring accuracy, Geographic Information System (GIS) compatibility, and enterprise software integration. It also emphasizes risk assessment, data analysis, and the application of AI to enhance anomaly detection. IEC TS 62446-3 [48] outlines required conditions for PV infrared thermography, offering guidelines for image interpretation, fault detection, and anomaly classification.

Inspection requirements must also consider the unique characteristics of each asset type. To clarify these differences, the following classification outlines the key attributes that influence data acquisition and processing across asset types:

- **Power Generation Systems**
  - **Photovoltaic Plants**: Large surface area and uniform asset characteristics. Data acquisition is highly repeatable with fixed distances and angles. Major defects can be identified during normal system operation. Detection of minor defects may require intrusive techniques such as electroluminescence.
  - **Wind Turbine Blades**: Medium surface area and uniform asset characteristics. High variability in the scale of defects. Detailed inspections require halting system operation, while major defects can be detected from higher distances and even with blades in motion.
  - **Hydroelectric Dam Walls**: Large surface area, with exposure conditions dependent on operational state. The submerged face of the walls poses additional challenges for data acquisition, not only due to the need for waterproof platforms and sensors, but also because of optical variations underwater.
- **Transmission and Distribution Systems**
  - **Power Lines and Structures**: Medium-sized assets installed at varying heights and spans, with high variability in shape – from linear elements (e.g., conductors and ground wires) to more complex structures (e.g., towers and poles). Inspections require adaptable platforms due to remote locations and safety constraints.
  - **HV Insulators and Other Components**: Generally small to tiny components, ranging from visibility markers to nuts and locking pins. These elements exhibit irregular, three-dimensional shapes with limited surface area exposure, requiring multiple viewing angles to ensure full coverage. Close-range data acquisition is critical for detecting subtle surface symptoms or missing parts.
  - **Substations and Transformers**: Compact yet complex assets composed of densely packed components. Real-time monitoring using technologies like infrared thermography (IRT) and ultraviolet (UV) sensors is effective for identifying thermal anomalies and partial discharges. Their fixed geometry makes them well-suited for permanent monitoring platforms.

These variations in asset size, shape, and operational context highlight the need for flexible, context-aware inspection strategies. Effective data acquisition must adapt to each asset's physical characteristics and inspection constraints, ensuring coverage and resolution are adequate for identifying relevant defects. A clear understanding of these characteristics is fundamental for designing automated inspection workflows that enhance reliability and safety while optimizing efficiency.

### C. PROCESS AUTOMATING

To effectively automate electrical grid asset inspections, a well-defined system decomposition is essential. Automation should be implemented progressively at each stage, determined by the operational and economic feasibility of the respective task. This approach ensures that the level of automation is optimized for each subsystem, balancing cost-efficiency with performance requirements.

Figure 11 outlines the full spectrum of grid inspections, from high-level asset management and inspection planning to asset quality reporting and maintenance operations. The assets to be inspected are defined by the management strategy, leading to inspection plans that outline, among other aspects, the resources to be used. This is followed by the data collection procedure, using technologies appropriate to each task. The collected data is then processed to identify the respective defects, resulting in a report on the quality and condition of the assets. This report guides potential replacements and repairs, ensuring the continuity and safety of the electrical system.

The most prominent bottlenecks occur during on-site data collection and analytical processing. Automating visual inspections improves efficiency and consistency by addressing the limitations of human inspections. Manual checks are time- and labor-intensive, require a skilled workforce, and are cognitively demanding. Inconsistent human judgments, influenced by fatigue and cognitive bias, can lead to errors and variability in decision thresholds [35].

Electric utility companies are increasingly collecting visual data during inspections to detect flaws and prioritize maintenance. While increasing the volume of data can raise the number of true positives, it does not necessarily improve sensitivity (recall) under normal testing conditions. Moreover, it introduces redundancy, which reduces efficiency when data is processed manually. The resulting data volume can further increase the bottleneck in human interpretation tasks.

Although the primary focus is on data processing, treating each stage as an isolated subsystem risks overlooking emergent properties [36] that arise from their interaction. The inspection process is not just a linear sequence of tasks but a dynamic system where decisions at each stage affect the others. For instance, decisions made at the management level influence data collection needs, while limitations in data acquisition can propagate upstream, affecting the accuracy of asset assessments. Therefore, the interface between these subsystems should not be rigid. An integrated approach, accounting for feedback loops and cross-domain dependencies, is essential to optimize grid operations and capture the full complexity of the inspection process.

Automation exists at multiple levels, each requiring varying degrees of procedural detail and standardization. Similar to the levels of autonomous driving (ranging from no automation to full autonomy) [49], electrical grid inspections can range from manual processes to fully autonomous systems.

At the basic level of automation, electrical grid inspections involve structuring and formalizing data acquisition procedures to ensure consistency when tasks are delegated to field operators or outsourced to third-party service providers. By defining objective inspection protocols, organizations can scale data collection without relying on specialized expertise at every step. On the higher end, fully autonomous data collection—such as UAVs autonomously navigating and capturing inspection data—requires translating the implicit knowledge of skilled inspectors into explicit, programmable rules. Skilled operators instinctively adjust camera angles to avoid glare, adapt to environmental conditions. These processes must be formalized to ensure safe navigation, account-

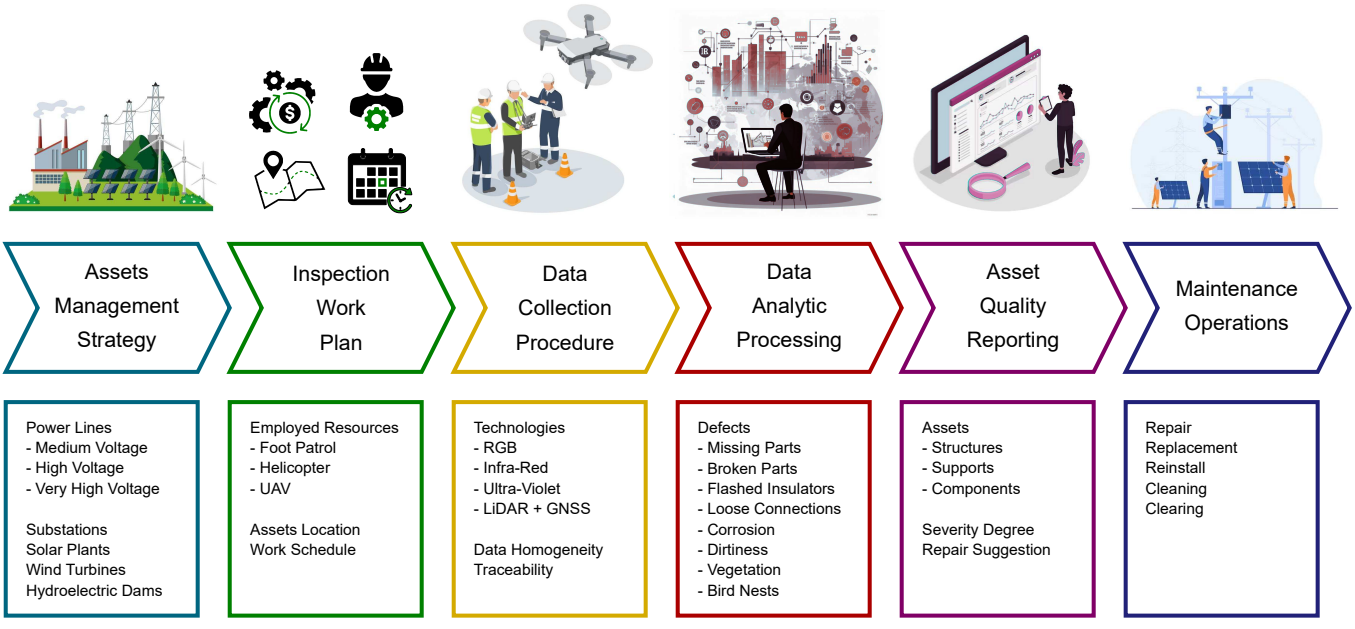



**FIGURE 11.** Overview of the current electrical grid inspection workflow. The assets to be inspected are determined by the management strategy, which leads to inspection plans detailing the required resources. Data collection follows, using appropriate technologies for each task. The data is then processed to identify defects, producing a report on asset quality and condition to guide potential replacements and repairs.

ing with obstacle avoidance and quick reactions to external factors like wind. Additionally, transitioning to high levels of automation requires systems to be robust against electromagnetic interference (EMI) and to comply with regulatory standards. For UAV-based inspections to be operationally viable and safely integrated into grid inspection workflows.

While the technical aspects of automation are crucial, the economic viability of automated systems is just as important. Automating the detection of frequent, high-impact defects is often a justifiable investment due to the significant benefits of early detection and reduced operational costs. However, automating rare or low-risk failure modes may not be cost-effective [50]. The high costs of developing, deploying, and maintaining automated solutions for low-impact anomalies may outweigh the benefits, making periodic manual inspections or targeted sampling a more practical option.

For example, in assets like PV plants, defects are more standardized and predictable. A limited set of failure modes can be identified and cataloged [48], which makes automation more feasible. This predictability, combined with the growing scale of PV plants, makes automation both operationally and economically viable. On the other hand, systems like electrical installations or turbine generators have a much greater variety of potential defects. The variability in dimensions, positions, and types of faults complicates automation. It's difficult to develop a comprehensive list of potential issues because electrical installations, for instance, have complex and dynamic conditions that vary significantly between installations, making it challenging to create a one-size-fits-all automated inspection system.

This increased variability raises both the technical and economic complexity of automation. Developing automated systems for such assets requires advanced sensing, perception, and decision-making capabilities, which significantly raises the cost of implementation. Moreover, because the potential defects are less predictable, the economic benefits of automating these inspections may not justify the high costs, especially when compared to more predictable and manageable systems like PV plants. Therefore, a careful balance must be struck between the costs of automation and its practical benefits. The economic and practical feasibility of automation depends largely on how predictable and standardized the defects are. For assets with more varied or complex defect patterns, a hybrid approach combining automation with human oversight may be a more viable solution.

## III. DATA ACQUISITION

This work defines data as visual information that can be represented in pixels or voxels, including visible, thermal, and depth data, as well as point clouds. This definition is grounded in the fact that all these data types fundamentally convey spatial information, in addition to amplitude information, which is crucial for interpretation and analysis in a visual context.

Data acquisition is structured as a sparse combination of three main independent dimensions: sensor technologies, platforms and access methods, and the inspected asset.

### A. TECHNOLOGIES AND SENSORS

Sensor technologies are categorized into two main principles: active and passive. Active sensors require an emitting component to generate energy, such as electromagnetic radiation, to capture visual data from the environment, as seen in X-ray sensors. Passive sensors, on the other hand, detect naturally occurring radiation, such as visible light or thermal radiation, without emitting energy. This subsection summarizes the underlying physics principles for each technology and sensor, followed by key challenges, limitations, and practical applications in electrical grid asset inspection.

Table 2 summarizes the main technologies and sensors used for automated electrical grid inspections. Sensor fusion data, such as RGB-D, is excluded since depth maps can be generated through various methods (e.g., stereoscopic, ToF) and do not constitute a standalone technology.

#### 1) Visible Light Imaging (VLI)

VLI captures scenes as perceived by the human eye, within the 400-700 nm range. Digital cameras use image sensors to convert light into electrical signals. The two main sensor types are CCD—which offers high image quality at high power consumption, and CMOS—which enables faster readout with lower power use, making it the modern standard. Each sensor pixel contains a photodiode that converts absorbed photons into an electrical charge via the photoelectric effect. This charge is then transformed into a voltage signal and digitized by an Analog-to-Digital Converter (ADC), where bit depth (e.g., 8-bit = 256 levels) determines intensity resolution.

Just like the human eye, visible cameras are passive sensors that rely on ambient light. Their performance is also reduced by atmospheric conditions such as fog, haze, smoke, heat waves, and smog, often limiting their applications to daytime and clear skies. Additional challenges that obscure details in VLI include glare and reflections from sunlight or artificial light sources on metallic or glossy surfaces, as well as low contrast between defects and the background.

VLI plays a crucial role in electrical grid inspections, providing high-resolution color images that help assess infrastructure integrity and detect anomalies indicative of potential failures. The wide range of defect detection applications can be grouped into categories such as surface anomalies (e.g., corrosion, rust, dirt, coating damage, cracks), shape deviations (e.g., broken, bent, or twisted assets), missing components (e.g., bolts or fittings), and misplacements (e.g., foreign objects such as bird nests or plastic bags in PL).

#### 2) Infra-Red Thermography (IRT)

IRT captures infrared radiation emitted by objects to visualize temperature variations. Since all objects above absolute zero emit IR radiation, thermal imaging provides a non-contact method for detecting heat patterns, making it useful for fault detection and diagnostics. IR images are typically monochrome, as standard sensors do not distinguish between different IR wavelengths. However, pseudo-color techniques enhance interpretation by mapping temperature to colors.

Thermal cameras use either cooled or uncooled detectors to capture IR radiation. Cooled detectors operate at cryogenic temperatures to prevent self-emission from overwhelming the sensor. This results in higher sensitivity and resolution but requires expensive, bulky, and energy-intensive cooling systems. In contrast, uncooled detectors function at ambient temperature. While they are more compact, cost-effective, and energy-efficient, they have lower resolution, slower response times, and must compensate for their own heat signature [56]. The radiant heat energy received by a thermographic camera is influenced by surface reflections and emissivity, which measures how effectively an object emits IR radiation, ranging from low values for polished metals to high values for rough surfaces. Emissivity is a critical factor, that can introduce errors in temperature measurements, but it can be compensated by using appropriate techniques [57]. Other factors include viewing angle, distance from the target, surface conditions (dirt, moisture, oxidation), atmospheric interference (dust, water vapor), and outdoor factors like wind-induced convective cooling. Understanding and compensating for these factors is crucial for precise thermal measurements [58].

Thermography is widely used in grid inspections to identify thermal anomalies that indicate potential failures. Since electrical components generate heat due to resistance, abnormal temperature patterns can reveal loose connections, corroded terminals, overloaded circuits, high-resistance points in electrical joints, busbars, and switchgear, as well as insulation degradation, moisture intrusion, and load imbalances in three-phase systems [59]. In PV systems, thermography can detect hotspots, bypass diode failures, and cell degradation [14].

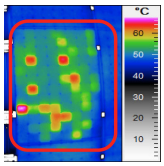
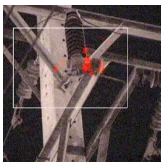
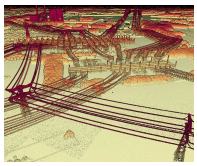
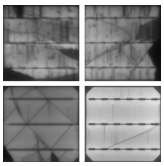
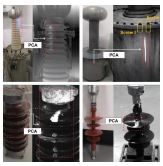
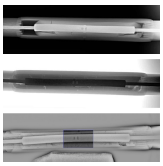
### 3) Ultra-Violet Imaging (UVI)

UVI is based on the detection of ultraviolet radiation emitted by ionized gas (plasma) during corona discharge. High voltage creates intense electric fields, which accelerate free electrons that ionize air molecules and emit UV light [60]. Passive UV cameras capture this radiation, converting it into visible signals that are tipically overlaid with visible light images to pinpoint the location and intensity of the corona discharge.

Corona discharge is a common phenomenon in high-voltage transmission systems due to concentrated electric fields. It can lead to power loss, noise, and radio frequency interference energy loss, equipment damage, safety hazards, and is also a significant cause of degradation in polymeric insulators. Solar-blind UV cameras, or day-time corona cameras are widely used in power lines, transformers, and high-voltage equipment to identify corona discharges before they lead to critical faults. These discharges typically occur at sharp edges or areas with high voltage gradients, such as around insulators, electrical contacts, and joints.

Contaminants on disc insulators further increase corona activity, accelerating material deterioration. Detecting these discharges is crucial for initiating maintenance procedures, such as water cleaning, to restore insulation performance and prevent failures [61]. Poorly maintained or degraded components, such as arcing contacts and damaged insulation, are also more susceptible to corona formation, which can compromise equipment reliability and safety. Early detection through UVI helps mitigate these risks, ensuring the longevity and efficiency of high-voltage infrastructure.

**TABLE 2. Overview of Visual Sensing Technologies for Electrical Grid Asset Inspection Tasks**

| **Principle** | Passive | | | Active | | | |
|---|---|---|---|---|---|---|---|
| **Band** | Visible | IR | UV | IR | | MSI / HSI | X-Ray |
| **Wavelength** | 400-700 *nm* | 8-14 $\mu m$ | 240-260 *nm* | 980 - 1600 *nm* | 1125-1175 *nm* | 400 - 1700 *nm* | 0.05 - 0.25 *nm* |
| **Technology** | Color Filtering | Thermal Irradiance Measurement | Corona Effect | Reflectance 3D Mapping | Luminescence Effect | Fluorescence Spectrum Analysis | X-Ray Penetration |
| **Sensors** | RGB Camera | IRT Camera | UVI Camera | LiDAR | InGaAs Camera Si CCD Camera | Hyperspectral Imaging Camera | X-Ray Scanner |
| **Data Structure** | 2D × 3 Ch | 2D × 1 Ch | 2D × 1 Ch | 3D × n[1] Ch | 2D × 1 Ch | 2D × n[2] Ch | 2D × 1 Ch |
| **Inspection Tasks** | Surface anomalies Shape deviations Missing components Misplaced parts | Overheating components Faulty connections Insulation failures Load imbalances | HV discharges Insulation breakdown | Infrastructure profiling Tower tilting Cable sagging Vegetation encroachment | PV cell cracks Isolated defects Series resistance Thermal history | Oil leakage Vegetation detection | Internal defects Mechanical attachments and fittings of PL Conductors |
| **Visual Samples** |  |  |  |  |  |  |  |
| **Reference** | [47] | [48] | [51] | [52] | [53] | [54] | [55] |

[1] Point clouds may include channels like RGB, reflectance, or have no additional attributes.
[2] Multispectral typically has 4-10 channels, while hyperspectral can have tens to hundreds.

#### 4) 3D Point Cloud (PC)

Point clouds provide a digital representation of physical environments by capturing discrete data points in a three-dimensional coordinate system. These are typically generated using LiDAR (Light Detection and Ranging), an active sensing technology that emits laser pulses and measures the time-of-flight (ToF) of reflected signals to determine distances with high precision. Alternatively, 3D point clouds can also be obtained passively through photogrammetry, which reconstructs 3D structures from multiple 2D images by identifying common feature points and applying geometric triangulation.

In electrical grid asset inspections, both methods present challenges. LiDAR requires precise calibration to maintain accuracy, while photogrammetry depends on stable camera positioning and sufficient image overlap. Additionally, large-scale inspections produce high-density point clouds, requiring substantial computational resources for processing it.

Despite these challenges, PCs play a crucial role in structural assessment by detecting deformations and misalignments in towers, conductors, and insulators. They are also used for clearance monitoring, ensuring safe distances between conductors, vegetation, and infrastructure [62].

#### 5) Luminescence

Luminescence refers to the emission of light from a material after it absorbs energy. In the context of PV inspections, the three main types of luminescence are: UV Fluorescence (UVF); Photoluminescence (PL); and Electroluminescence (EL). UVF occurs when a material emits light after exposure to ultraviolet (UV) light. PL happens when a material absorbs photons, typically from sunlight, LEDs, or lasers, and re-emits them. EL occurs when an electric current is applied to a material, causing it to emit light [13].

Despite their benefits, luminescence techniques face several limitations. UVF depends on external UV sources and ambient lighting. PL and EL require controlled environments—typically labs or night-time conditions. EL also needs electrical biasing and is sensitive to thermal effects, complicating outdoor use. Additionally, the emitted luminescence signal is often weak, comparing with strong background light, making outdoor defect detection difficult. However, optical filters, specialized sensors, and advanced analysis methods can help isolate the signal [63]. Non-destructive methods like IR and EL imaging help detect hidden defects that impact module efficiency. IR imaging identifies disconnected or damaged cells by capturing emitted infrared radiation but may miss fine defects like microcracks due to resolution limits. EL offers higher resolution and can reveal early-stage defects, but it requires electrical biasing and is sensitive to thermal effects, making it more suitable for controlled environments.

Although luminescence techniques can be applied to material quality analysis in industries like materials and semiconductor manufacturing, this work focuses on their use for PV inspections. For instance, EL imaging detects microcracks and areas of increased series resistance in PV modules, while PL helps assess semiconductor quality and thermal history.

#### 6) Multi and Hyper-Spectral Imaging (MSI/HSI)

MSI and HSI capture detailed spectral information beyond the visible spectrum. MSI records images in a limited number of discrete spectral bands (typically $3 \sim 10$), which are selected based on specific applications, making it effective for broad material classification. HSI, on the other hand, captures a continuous spectrum across hundreds of narrow bands, enabling finer spectral resolution and precise material identification.

One key application of MSI is the Normalized Difference Vegetation Index (NDVI), which assesses vegetation health by analyzing the difference between near-infrared (NIR) and red light reflectance. Healthy vegetation reflects more NIR and absorbs red light, while stressed vegetation reflects more red light, resulting in lower NDVI values. By generating NDVI maps, inspectors can identify overgrown or unhealthy vegetation near transmission lines, reducing the risk of faults, outages, and wildfires. Hyperspectral imaging (HSI) has a wide range of applications, from agriculture and food processing to mineralogy and medical diagnostics [64].

In electrical grid inspections, MSI is useful for large-scale monitoring, such as detecting vegetation encroachment near power lines [65], while HSI provides deeper material analysis, such as identifying insulation degradation [66] or detecting oil leaks in transformers [54]. Oil leakage can lead to temperature anomalies and insulation failures, posing significant risks to grid reliability. Recent advances, such as ultraviolet-induced hyperspectral imaging (UV-induced HSI), improve oil leak detection by analyzing fluorescence and reflectance intensity under UV light, enhancing defect detection rates.

#### 7) X-Ray Radiography

X-Ray Radiography is a non-destructive testing (NDT) method that enables internal inspection of materials and components without causing damage. While commonly associated with medical diagnostics, it also plays a crucial role in industrial and asset inspections. In electrical grid maintenance, X-ray technology is used to examine the internal structure of transformers, circuit breakers, and cables, revealing faults that may not be visible externally.

This technique relies on high-energy radiation passing through materials, with denser regions absorbing more X-rays and creating contrasting images. These images help detect internal defects such as cracks, voids, gas buildup, or insulation degradation, providing valuable insights for failure prediction and preventive maintenance. Unlike other inspection methods, X-ray imaging allows for internal assessments without disassembling equipment, reducing downtime and improving efficiency. It can also help identify material inconsistencies or structural defects in composite components.

Despite its advantages, X-ray radiography has limitations, including high operational costs, strict safety requirements due to radiation exposure, and challenges in accessing certain components. However, the development of portable X-ray systems has expanded its applications, enabling on-site inspections of power lines and cables in difficult-to-reach areas [67].

### B. MEANS AND PLATFORMS

Historically, electrical grid inspections relied on manual, labor-intensive methods, with field personnel physically assessing asset conditions. While effective at small scales, these processes are time-consuming, costly, and pose safety risks. Technological advancements have introduced automated inspection means, including the use of drones, and robots. Computer vision has emerged as a powerful tool for automating the inspection process by enabling precise, remote fault detection.

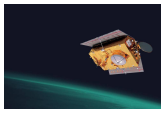
Table 3 summarizes the advantages and disadvantages of the main platform categories used in automated grid asset inspections. For a fair comparison of inspection platforms, manned systems are considered independently of operator expertise. It is assumed that data is systematically collected without real-time condition assessment. While expert operators may reduce irrelevant data, this selective approach risks missing faults not identified during fieldwork.

#### 1) Stationary Platforms

Stationary Platforms refer to fixed structures that support various sensors for grid asset inspection. While satellites are not strictly stationary, they share key traits with fixed platforms, such as continuous data collection from pre-defined observation points, making them functionally comparable in this context. While mobile platforms are limited by their energy autonomy, fixed platforms benefit from a continuous power supply, allowing uninterrupted data acquisition and real-time monitoring of grid assets.

The main differences between satellites and local structures are image resolution and coverage. A key factor in decision-making is the trade-off between monitoring area and required detail. Sensor-to-asset distance affects both resolution and coverage. High-resolution satellites offer 0.25–0.5 m/pixel [77], while local structures achieve higher resolution using relatively inexpensive sensors. Satellites require costly sen-

**TABLE 3. Overview of Means and Platforms used for Electrical Grid Asset Inspection Tasks**

| | Platform | Illustration | Ref | Advantages | Disadvantages |
|---|---|---|---|---|---|
| Stationary | Structure |  | [68] | Continuous and real-time monitoring<br>High resolution using standard sensors<br>Stable data collection<br>Low cost for small-scale areas | Limited coverage area<br>Maintenance requirements<br>Requires installation<br>Expensive to expand coverage |
| | Satellite |  | [69] | Global coverage<br>Minimal maintenance<br>Cost-effective for large areas<br>On-demand service | Low resolution<br>Lag in data acquisition<br>High initial cost<br>Dependency on weather conditions |
| Manned | Foot Patrol |  | [70] | Adaptability to diverse environments<br>Cost-effective for dense asset areas<br>Flexibility on the data detail<br>Access to hard-to-reach areas | Limited coverage for widespread assets<br>Safety risks at heights or live assets<br>Labor-intensive for large-scale inspections<br>Slow inspection speed over long distances |
| | Road Vehicle |  | [71] | Faster data collection for distributed assets<br>Long-range capability and wide-area coverage<br>Supports heavy equipment and multiple sensors<br>Enables frequent and passive data acquisition | Limited to assets near drivable paths<br>Cannot inspect elevated structures effectively<br>Lower flexibility in navigating obstacles<br>Slower than aerial methods |
| | Helicopter |  | [72] | Rapid inspection over large distances<br>Access to remote and difficult terrain<br>High-resolution imaging from aerial perspective<br>High payload capacity of sensing equipment | High operational and maintenance costs<br>Weather-dependent flight conditions<br>Requires specialized pilots and permissions<br>Limited low-altitude stability for precise data |
| Unmanned | Climbing Robot |  | [73] | Access to live power lines<br>Enables detailed conductor inspections<br>Reduces human exposure to high-risk areas<br>Can operate autonomously in certain cases | Limited to power lines or similar structures<br>Complex deployment and retrieval<br>Slow and limited mobility<br>Potential wear on conductors over time |
| | Multi-Rotor |  | [74] | High maneuverability in tight spaces<br>Provides detailed aerial imaging<br>Access to hard-to-reach areas<br>Rapid deployment with minimal setup | Limited flight time and range<br>Weather-sensitive performance<br>Requires skilled operator<br>Low payload capacity |
| | Fixed-Wing |  | [75] | Efficient for large area coverage<br>Long flight endurance<br>High-speed operation over long distances<br>Cost-effective large-scale monitoring | Requires takeoff/landing space<br>Less stable for low-altitude<br>Harder to retrieve in case of failure<br>Require precise sensors to deal with motion |
| | Hybrid VTOL |  | [76] | Combines vertical and horizontal flight<br>Provides coverage over large areas<br>Capable of vertical takeoff and landing<br>More flexible deployment options | More complex system than multi-rotors<br>Higher cost compared to quadcopters<br>Requires specialized training and operation<br>Increased maintenance demands |

sors but offer near-global coverage. Local structures need additional installations to expand range. Though cheaper initially, their expansion costs are high. Due to satellites' astronomical costs, outsourcing satellite-based inspection to specialized providers is often more practical. Another difference relates to data acquisition frequency. Local platforms are often proprietary, while satellites are essentially a rental service. Depending on the contracted service, this might introduce a delay in data acquisition, preventing real-time monitoring.

While continuous monitoring may seem useful, equipping every asset with fixed monitoring equipment is impractical. The operational costs of maintaining such a system would likely exceed the costs of maintaining the electrical grid assets themselves. However, continuous monitoring becomes more justifiable for critical assets, such as transformers and substations, due to their higher risk and potential impact on grid reliability. For most assets, which degrade slowly and do not require immediate intervention, mobile platforms provide a more economically practical solution.

#### 2) Manned Means

Manned Means are a subset of mobile platforms where onboard human operators control the sensor's position relative to the asset. This work categorizes three types of manned platforms: human operators handling sensors directly, sensors mounted on road vehicles, and attached to aerial vehicles.

Foot patrols are among the most cost-effective, easily deployable inspection methods, particularly in urban environments, offering high adaptability. Their flexibility allows operation across a wide range of scenarios, from collecting data to inspecting electrical assets in indoor facilities and outdoor substations. Additionally, they can navigate dense vegetation, obtain closer viewpoints of supporting structure bases, and achieve a bottom-up perspective on higher assets, enabling more detailed data capture and, in specific cases, more accurate defect identification compared to aerial methods [78].

However, high exposure to safety risks, along with their low mobility speed and limited range, makes them impractical for inspecting large-scale assets such as transmission and distribution grids, where other means are often more efficient.

Road vehicles serve as mobile inspection platforms by carrying onboard sensor systems, enabling efficient data collection along accessible routes. They facilitate faster inspections than foot patrols and can operate passively by integrating sensors into existing fleet vehicles without intrusive modifications [71]. However, road-based platforms are limited to assets near drivable paths, typically within distribution grids.

Helicopters are the primary manned aerial platform for electrical grid inspections, valued for their maneuverability. Their high payload capacity allows the integration of multiple or heavier sensing systems for comprehensive inspections in a single flight. They provide rapid coverage of large areas and enable access to remote locations and mountainous terrain, making them ideal for long HV transmission lines. However, helicopter inspections involve high operational costs, which can reduce efficiency in some environments.

While manned means remain essential in specific contexts, such as detailed inspections or navigating complex environments, their scalability and cost-efficiency are limited compared to unmanned platforms. Although helicopters offer greater reach than road vehicles, their cost and logistical challenges make them viable mainly for critical or high-priority areas, rather than for frequent routine inspections.

#### 3) Unmanned Means

Unmanned means are mobile platforms where operators remotely control the sensor's position relative to the asset. UAVs (Unmanned Aerial Vehicles) are the popular option, but climbing robots offer unique advantages as well. Unmanned platforms are lightweight, cost-efficient, and have lower regulatory requirements, but they are constrained by limited autonomy, reduced payload capacity for sensor equipment, and limited range for accessing distant locations. The trade-offs between their mobility, endurance, and data collection capabilities offer diverse solutions for specific inspection needs.

Climbing robots are designed to navigate live PLs, offering close-range inspections of cable conductors, enabling detailed data collection. For instance, LineSpyX [67] is equipped with a X-ray sensor to inspect the condition of the conductor core. Robots like LineScout [73], with a telescopic frame, can overcome common obstacles found on the grid, broadening their inspection capabilities. LineRanger [79] further extends this by enabling other maintenance tasks on the power lines. However, their mobility is limited as they are restricted to the power lines and require specialized deployment methods.

Multi-rotor drones ability's to hover offers precise maneuverability, which makes them suitable for detailed data collection at varying angles. They are widely used for static asset assessments but are limited by short flight duration. To address UAV flight autonomy challenges caused by load constraints that limit battery capacity, some studies explore harvesting energy from high-voltage power lines to recharge batteries using current transformers [80], [81]. However, the added recharging system increases weight, potentially limiting sensor payload capacity, speed, and maneuverability.

Fixed-wing drones can cover larger areas due to better flight aerodynamics and higher speeds, making them efficient for inspecting long infrastructure extensions. However, their inability to hover requires a launch and landing area, and the constant motion may introduce noise in sensor data.

Hybrid vertical takeoff and landing (VTOL) drones combine the advantages of both multi-rotor and fixed-wing UAVs. They can take off and land vertically like a multi-rotor but transition into efficient forward flight like a fixed-wing drone. This versatility allows them to perform both localized and large-area inspections, though they tend to be more complex and expensive than single-mode UAVs.

Unmanned means are becoming increasingly prominent in grid inspection due to their adaptability, autonomous navigation potential, and ability to access hazardous environments. However, their effectiveness depends on regulatory constraints, operational logistics, and energy limitations.

### C. DEFECTS SCALE AND INSPECTION DISTANCE

Selecting sensor modalities for specific inspections tasks is relatively straightforward. For example, VLI for surface defects, IRT for resistive heating, or LiDAR for geometric measurements. However, an often overlooked factor in optimizing data acquisition is the relationship between the inspection platform, its operational distance, and the size of the defects being targeted. This interplay directly influences the resolution, accuracy, and overall effectiveness of condition monitoring and subsequent analysis. Fig. 12 illustrates the operational inspection distances associated with different platforms.

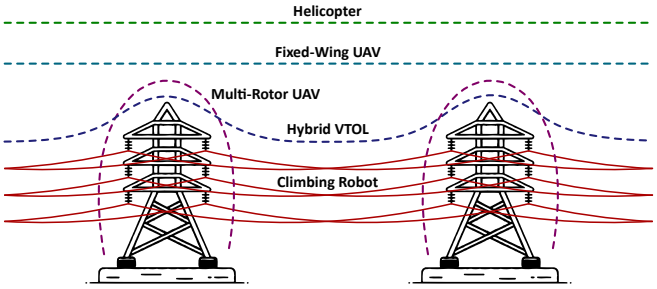



**FIGURE 12.** Overview of operational inspection distances for different platforms, adapted from [82]. Helicopters and fixed-wing UAVs require greater safety distances due to low maneuverability. VTOLs can operate closer to power lines and structures, while multi-rotor UAVs excel in localized inspections. Climbing robots are restricted to cable navigation.

Close-range imagery, typically used to assess insulator shell conditions, is unsuitable for detecting tower misalignment or cable sagging. Conversely, aerial inspections conducted at high speeds are well-suited for identifying shattered insulator discs but pose challenges when detecting missing nuts or broken conductor strands in PLs. Table 4 categorizes key grid inspection tasks based on defect scale and inspection distance.

**TABLE 4.** Overview of Inspections Distances and Tasks

| Defect Scale | Large | Medium | Small |
|---|---|---|---|
| Asset Dimension | meters | centimeters | millimeters |
| Suitable Platform | Helicopter<br>Fixed-wing UAV | Multi-rotor UAV<br>Hybrid VTOL UAV | Climbing robot<br>Hybrid VTOL UAV |
| Inspection Tasks | Leaning Structures<br>Lines Sagging<br>Vegetation Clearance | Insulators Damage<br>Visibility Markers<br>Attachments and Fittings | Small Cracks<br>Nuts and Pins<br>Broken Wire Strands |
| Illustration |  |  |  |
| Reference | [83] | [47] | [84], [85] |

Grouping inspection tasks by distance is crucial as it directly impacts sensor selection, platform choice, data quality, and overall operational efficiency. Once data is acquired, different annotation procedures can be applied depending on the specific inspection objectives. However, rather than constraining data processing methods to fit suboptimal datasets, acquisition strategies should be designed to support the objectives of downstream analysis. Therefore, data collection strategies must align with the entire inspection pipeline, ensuring that acquisition procedures optimize both defect detection accuracy and efficient data utilization.

## IV. DATA PROCESSING

In this work, we limited the scope of visual data processing to Deep Learning (DL) methods, focusing not only on current approaches applied to electrical grid asset inspections but also on emerging paradigms expected to be adopted in the near future. To understand the complex combinatorial set of modular solutions, this section briefly describes the evolution of DL, highlighting the main contributions to the field. It characterizes the dimensions and domains of the components required to construct a DL model and reveals key research topics that could overcome current limitations and elevate automated visual inspections to the next level.

### A. DEEP VISION EVOLUTION

Deep learning for computer vision has progressed through distinct stages, from Convolutional Neural Networks (CNNs) to Vision Transformers (ViTs) and, more recently, Large Vision Models (LVMs). These advancements have significantly improved image recognition tasks, driving innovation across various domains [86]. The origins of ML trace back to the 1960s with the perceptron model [87], while the potential of backpropagation in multilayer architectures emerged in the 1970s and 1980s [88]. Although CNNs started being used in the late 1990s [89], they only gained widespread adoption in the 2010s. The 2000s introduced unsupervised learning techniques [90], which, combined with the acceleration provided by Graphics Processing Units (GPUs) [91], fueled large-scale DL experiments. By the 2020s, Transformer architectures began reshaping CV, leading to ViTs and eventually LVMs, which leverage multimodal learning to achieve zero-shot generalization capabilities. In summary, as illustrated in Fig. 13, Digital Image Processing (DIP), which relied on explicit visual feature extraction and classification, evolved into Multi-Layer Perceptrons (MLPs), enabling learned classification of these features. CNNs transformed feature extraction and formed the backbone of object detection models, ViTs introduced self-attention for capturing long-range dependencies, and LVMs unlocked new possibilities through multimodal prompting and large-scale training.




**FIGURE 13.** Overview of DL evolution applied to computer vision

#### 1) Convolutional Neural Networks

CNNs have been foundational in CV tasks, such as image classification and object detection. Their ability to hierarchically extract features has been extensively validated in tasks handling structured data distributions, such as images. A typical CNN model has two main components: a feature extractor, composed of convolutional layers, and a classifier, usually consisting of dense layers. As the network advances, spatial dimensions shrink while depth increases, transforming a high-dimensional matrix into a reduced vector.

The history of CNNs began in 1998 with the introduction of LeNet-5 [89], a pioneering architecture composed of 5 layers (3 *Conv* + 2 *Dense*). In 2012, AlexNet [92], marked a significant leap in computer vision by classifying 1.2 million mid-resolution images into 1,000 categories with 37.5% top-1 error rate, and winning the ImageNet competition. VGG16 [93], developed in 2014, featured a deeper architecture, housing 138 M parameters across 13 *Conv* and 3 *Dense* layers. That same year, the GoogleNet [94] introduced the inception module, which captures multiscale features through parallel convolutions and pooling. This enabled a 22-layer NN with only 13 M weights, improving accuracy with fewer parameters.

ResNet [95], introduced by Microsoft in 2015, revolutionized DL by introducing residual connections, addressing the vanishing gradient problem in deep networks, and enabling effective training of NN with hundreds of layers. In 2017, Xception [96] replaced Inception modules with depthwise separable convolutions. DenseNet [97] set a new SotA by replacing sequential residual connections with dense ones, reducing the need for deep networks and achieving better results with fewer parameters. MobileNet [98] focused on efficient parameter scaling for mobile and embedded applications.

In 2020, EfficientNet [99] established new benchmarks with the introduction of an efficient compound scaling coefficient. Acknowledging that NN scaling occurs in three dimensions (image size, width, and depth), they study the optimal trade-off between these dimensions, resulting in better architectural choices. ConvNeXt [100] further improved the SotA by combining several previously introduced concepts. In practice, it involved a grid search of hyperparameters, including regularization, optimizer, and activation, incorporating those reported in recent vision transformer models.

### 2) Vision Transformers

Transformers marked a significant shift in DL methodologies. Originally developed for Natural Language Processing (NLP), the Transformer architecture outperformed traditional Recurrent Neural Networks (RNNs), which struggled to capture long-range dependencies. NLP has progressed from simple bag-of-words models [101] to more advanced word embeddings [102], which represent words as continuous vectors capturing semantic relationships. Early models, such as RNNs [103], and their improved versions, Long Short-Term Memory (LSTM) [104] networks, were designed to process sequential data but still struggled with capturing long-range dependencies. The paradigm shift occurred with the introduction of the Transformer architecture [105], which replaced recurrence with self-attention, enabling efficient parallel processing and improved context modeling. Self-attention computes similarity scores between input embeddings to capture long-range dependencies. This mechanism was later extended [106] to incorporate relative positional information.

This innovation enabled language models like BERT [107], which used bidirectional encoder representations to generate contextualized embeddings, achieving state-of-the-art results in NLP tasks like question answering and sentiment analysis.

Vision Transformers emerged in 2020 [108], introducing a novel approach to image analysis by dividing images into fixed-size patches (e.g., 16x16), embedding them as tokens, and applying multi-head self-attention mechanisms, similar to how words are processed in NLP. Unlike CNNs, which rely on local feature hierarchies, ViTs capture global dependencies through self-attention. Google's ViT [108] outperformed CNNs on the ImageNet benchmark but required pretraining on 300 million images instead of the standard 1.2 million. Later that year, Meta introduced the Data-efficient Image Transformer (DeiT) [109] to address the data-hungry nature of ViTs. DeiT incorporated a teacher-student training paradigm, where a CNN encoder generated a knowledge distillation token to guide the Transformer student in learning local features representations. This improved ViT efficiency, enhanced generalization, and reduced reliance on large datasets, making them more practical for real-world applications.

In 2021, the SWin Transformer [110] improved the ViT architecture by reintroducing ConvNet priors through the Shifted Windows scheme. This approach divided the image into non-overlapping windows and applied self-attention within each window, while shifting the windows across layers to capture long-range dependencies. The modification enhanced computational efficiency and scalability for high-resolution images, setting new benchmarks in CV tasks.

### 3) Large Vision Models & Multimodal LLMs

Foundation models [111] advanced AI by integrating modalities like audio and image, enabling sophisticated contextual reasoning and task generalization. Vision-language models have played a key role in pushing the boundaries of CV.

CLIP (Contrastive Language-Image Pretraining) [112], introduced by OpenAI in 2021, uses contrastive learning to map images to textual descriptions, enabling zero-shot capabilities in tasks like classification and detection. This approach allows it to be applied across various tasks without task-specific training. DALL-E [113], also developed by OpenAI in 2021, extends CLIP's capabilities by generating images from textual descriptions, enabling both text-to-image and image-to-text transformations for creative applications.

Flamingo [114], introduced by DeepMind in 2022, excels in few-shot learning, processing both images and text to quickly adapt to new tasks with minimal examples, making it suitable for dynamic scenarios. PaLI (Pathways Language and Image) [115], released by Google Research in 2022, combines vision and language understanding, demonstrating strong performance across diverse tasks and highlighting the power of cross-modal learning in AI.

SAM (Segment Anything Model) [116], developed by Meta in 2023, focuses on class-agnostic image segmentation. Trained on over 1 billion masks from 11 million images, it sets new benchmarks for flexibility and zero-shot performance across tasks. Its promptable design adapts easily to new scenarios, often outperforming prior models. The SAM model and its dataset (SA-1B) are publicly available, fostering further research in foundation models for CV.

In August 2023, Alibaba introduced Qwen-VL [117], a vision-language model for tasks like text reading, localization, image captioning, and question answering. In September 2023, OpenAI released GPT-4V [118], a MLLM that combines vision and language understanding.Meta released Llama 3 [119] in November 2024, a MLLM with 405 billion parameters, excelling in tasks like image, video, and speech recognition. In December 2024, Google DeepMind published PaliGemma 2 [120], an upgraded LVM from the Gemma 2 family, excelling in tasks such as table structuring and molecular structure recognition. Florence-VL [121], released by Microsoft Research in December 2024, builds upon the Florence-2 model by integrating enriched visual features through a novel Depth-Breadth Fusion architecture. Florence-VL excels in various multimodal tasks, including OCR, Visual Question Answering (VQA), and chart analysis.

Despite the remarkable performance of Multimodal Large Language Models (MLLMs) in generic tasks, their application in industry-specific contexts, such as electrical grid asset inspection, remains limited. Generic training datasets, like ImageNet [122] or MS-COCO [123], fail to capture the nuanced characteristics of industrial images, which often contain repetitive patterns, subtle defects, or low-contrast regions.

### B. ALGORITHMS AND ARCHITECTURES

As introduced earlier in Fig. 5, the algorithmic dimension is the most complex within the set of dimensions that constrict the *Model*, the core of the automated inspection system. Categorizing deep learning development is inherently challenging due to the variability in existing methodologies. This study introduces a structured yet concise taxonomy that organizes key dimensions within a coherent framework at Table 5.

While the table format provides a structured overview, it may not fully capture the overlapping and interdependent relationships among these terms. The combinations of subdimensions within this domain are inherently sparse, shaped by both dependencies and incompatibilities. For instance, contrastive learning can be either supervised or self-supervised [124], leading to distinct loss functions [125], whose applicability depends on the chosen learning paradigm.

An example of the blurred boundaries between subsystems, as discussed in Section II-C, is the data handling stage. Preprocessing and human-driven decisions tasks, such as training data selection and labeling, can be classified under the *Data* domain rather than the *Algorithm* domain. In either case, it is crucial not to overlook or neglect these stages, as their impact can propagate throughout the model design, ultimately limiting inspection performance.

Most publications, especially in domains like electrical grid asset inspection, tend to focus on the core stages of deep learning development, outlined in Table 5, NN architectural refinement and optimization of learning strategies. This model-centric approach has been effective in pushing the boundaries of deep learning models. However, the critical role of data is often underestimated [37], leading to developments that may not fully align with industry expectations.

**TABLE 5. Sub-Dimensions of Deep Learning Model Development**

| Stage | Process | Specification |
|---|---|---|
| **Data Handling** | Collection | Sourcing, Diversity, Splitting, Class Balancing |
| | Labeling | Curating, Spatial Granularity, Semantic Hierarchy |
| | Cleaning | Noise, Outliers, Missing Data Fields |
| **Preprocessing** | Normalization | Dataset, Batch, Block, Layer |
| | Augmentation | Geometric Transforms, Color Jittering, Mixup |
| | Synthesization | GANs, Diffusion Models |
| **Architectural Design** | Blocks | MLP, CNN, RNN, Auto-Encoder, Transformer |
| | Layers | Conv, Dense, Flatten, Pooling, etc. |
| | Size | Width, Depth, Parameter Count |
| | Connections | Residual, Skip, Dense Connections |
| | Mechanisms | Self-Attention, Gating |
| **Learning Strategy** | Paradigm | Supervised, Unsupervised, Semi-Supervised |
| | Loss Function | Cross-Entropy, Focal, Contrastive, Adversarial |
| | Optimization | Learning Rate, Batch Size, Momentum |
| | Regularization | Dropout, Weight Decay, Loss Penalty |
| **Inference & Deployment** | Compression | Pruning, Quantization, Distillation |
| | Optimization | Latency, Few-Shots Learning |
| | Environments | Edge Devices, Cloud Platforms, Mobile |
| **Evaluation & Benchmarking** | Metrics | Multi-Threshold Metrics, Single-Threshold Metrics |
| | Error Analysis | Confusion Matrix, Misclassification, Explainability |
| | Robustness | Cross-Validation, Domain Shift, Ablation, Attacks |

For instance, while standardizing benchmark metrics and testing datasets is essential for evaluating model progress, there is a risk of over-optimizing these factors, especially when datasets represent only a small sample of real-world diversity. As DL models evolve and become more powerful, it becomes increasingly important to shift focus toward the quality of the data and the robustness of model performance.

### C. EMERGING METHODOLOGIES

The AI landscape is buzzing with new learning paradigms and concepts, each aiming to push the boundaries of generalization. New forms of transfer learning, such as few- and zero-shot learning, seek to enable models to adapt to new tasks with few to zero labeled data. However, as models continue to grow in both parameter size and data consumption, they increasingly rely on vast amounts of Internet-sourced training data. This raises important questions about potential data overlap, as models may sometimes be evaluated on data similar to what they have seen during training [126], which complicates assessments of their true generalization capability.

Model development is shifting from learning from data to learning how to learn, a concept known as meta-learning [127]. Variants include Gradient-Based Hyperparameter Optimization [128], Neural Architecture Search (NAS) [129], and Curriculum Learning [130], which progressively trains models from simpler to more complex tasks. The latest trend in this evolution is the rise of AI agentic workflows, emphasizing autonomous decision-making and adaptability.

Generative AI has accelerated software development across domains by enabling rapid prototyping and fast iteration. This shift has made experimentation the new engine of innovation. However, as model development becomes increasingly frictionless, evaluation is emerging as the new bottleneck. Robust assessment criteria are evolving to keep pace with growing model complexity, and ensuring safety has become a critical concern. Without careful evaluation, there is a risk of deploying models with harmful or unintended behaviors [131].

### 1) Data-Centric AI

Data-Centric AI is the discipline of systematically engineering data to build successful AI systems. It focuses on improving data quality, diversity, and labeling strategies to enhance model performance, rather than solely optimizing training hyperparameters and model architectures [132].

As illustrated in Fig. 14, this paradigm introduces an additional iterative loop in DL development, which reflects the interdependence between subsystems. By aligning data assets and curation processes with overall system objectives, it mitigates the risk of overfitting to suboptimal, or biased datasets and enables better generalization in real-world conditions.

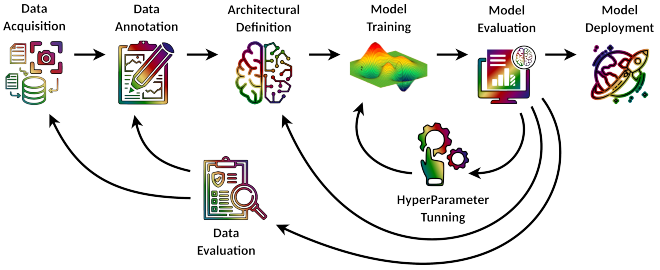



**FIGURE 14. Overview of model-centric and data-centric approaches. Model-centric feedback loops involve hyperparameter optimization, which may eventually expand to include architectural definition. The data-centric approach introduces an additional feedback loop that extends to earlier stages of model development, aiming to influence not only data annotations but eventually data acquisition as well.**

Over the past few decades, Andrew Ng [133] established new benchmarks for DL across various domains, including CV. Back in the 2010s, the premise that with enough data, the training algorithm would eventually encode relevant similarities and identify patterns within each data class was widely accepted. While DL revolutionized the signal processing field, the growing demand for data, revealed that more data does not necessarily mean more diversity. Despite DL models showing robustness to noisy labels, particularly with large datasets, clean labels consistently yield better performance [134]. Its recognized now that, in industries with limited datasets, the focus should shift from big data to high-quality data, since even a modest collection of meticulously curated examples with consistent and accurate annotations can effectively guide NNs toward achieving desired learning objectives [135].

Research efforts focused on data quality have revealed millions of labeling errors [136] in popular publicly available datasets, such as MNIST and ImageNet. By using probabilistic modeling and uncertainty estimation to detect potential label issues, CleanLab [137] enables the correction of these errors, thereby improving the overall quality of the dataset.

Another emerging methodology related to data-centric AI is Active Learning (AL), where the model selectively queries the most informative data points for labeling. The goal is to minimize the labeled data required to achieve high performance by focusing on the most valuable examples. ActiveLab [138] demonstrates this concept by automatically identifying when it is more efficient to re-label existing examples rather than labeling new ones, optimizing the overall labeling process.

### 2) AI Agentic Workflows

AI agentic workflows refer to automated, intelligent processes where systems actively take on decision-making, task execution, and adaptation roles. The term "agentic" emphasizes that the AI system not only responds to inputs but actively makes decisions and optimizes workflows. Here, workflows represent structured sequences of tasks that the AI system automates or enhances. Agentic AI represents a new abstraction layer in AI technology, connecting the capabilities of large models and infrastructure with high-level use cases. In this emerging five-level stack, agentic systems sit just below the top application layer and above foundational models, cloud infrastructure, and semiconductors [139].

Agentic AI design patterns—such as reflection, tool use, planning, and multi-agent collaboration—are central to building more capable and autonomous systems. By enabling iterative workflows, agentic AI allows LLMs to plan, revise, and interact with tools or agents, achieving significantly better performance than linear, zero-shot prompting.

**Reflection** is a simple yet powerful design pattern that enhances LLM output by prompting the model to critique and revise its own responses. Instead of producing a final answer in one step, the model generates an output, evaluates it for issues and revises it. This loop can be further improved using external tools or collaboration, and has shown significant gains across tasks like coding and question answering [140].

**Tool Use** enables LLMs to access external functions to retrieve information, perform actions, or manipulate data, extending their capabilities beyond pre-trained knowledge. Rather than relying solely on internal reasoning, the model detects when a tool is needed, invokes it, and incorporates the results into its reasoning [141].

**Planning** is a design pattern where a language model determines the sequence of steps to complete a complex task. Instead of following predefined instructions, the agent decomposes the problem, selects tools, and defines the execution order. This enables agents to reason through and organize actions for tasks like object detection or pose estimation [142].

**Multi-Agent Collaboration** mirrors collaborative human workflows, where contributors bring different expertise. Frameworks like AutoGen [143] support designing such systems. By enabling task decomposition and parallel execution, this pattern promotes specialization and efficiency, though coordination challenges may arise [144].

Agentic workflows have the potential to transform electrical grid asset inspection by enabling systems to move beyond static anomaly detection toward autonomous, context-aware decision-making. For instance, an agentic system could identify uncertain defects, assess confidence levels, and reacquire data from alternative viewpoints or modalities. However, deploying such workflows raises challenges, including ensuring safety and robustness, aligning decisions with human expectations and regulations, and managing real-time planning under resource constraints. Despite these hurdles, agentic AI presents a promising path toward more resilient, efficient, and intelligent inspection systems.

## V. APPLICATION SCENARIOS

This section presents a review of selected research works conducted by other authors, providing comprehensive coverage of the main dimensions previously discussed—inspected assets, platforms and means, technologies and sensors, and algorithms and architectures—without being excessively exhaustive. The review focuses on complementary sub-dimensions such as task description, data annotations, and model performance, offering a detailed descriptive analysis of each work, complemented by a succinct comparative analysis in table format that includes quantitative performance results.

### A. POWER GENERATION SYSTEMS

The electrical assets in power generation systems, considered in this review, such as PhotoVoltaic (PV) panels, Wind Turbine Blades (WTB), and Hydroelectric Dam Walls (HDW), share common characteristics that make them ideal targets for automated visual inspections. These assets are often massive in size, with extensive surface areas that cannot be captured in a single image shot. This requires complex acquisition processes, where images must be collected from multiple positions while maintaining consistent distances and angles relative to the asset. Furthermore, effective inspection demands an ID tracking system to correlate the images with respective parts of the asset, ensuring complete coverage and supporting following maintenance actions.

#### 1) Photovoltaic Modules and Cells

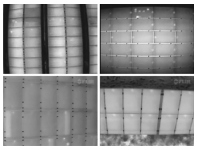
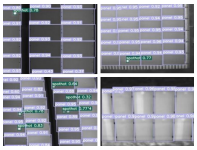
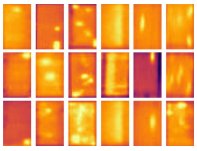
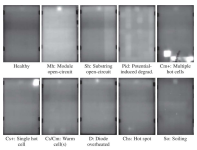
The most commonly used platform for PV plants inspection is a UAV platform equipped with thermographic cameras. This setup generates infrared thermal (IRT) data with annotations that may vary according to the task description.

The IEC TS 62446-3 standard [48] specifies the conditions under which IRT PV inspections should be carried out to ensure accurate and reliable results, such as ambient temperature, irradiance levels, and viewing angles. It provides guidelines for interpreting thermal images and identifying different types of faults based on temperature variations and thermal patterns. The standard includes criteria for classifying and prioritizing detected anomalies for maintenance actions. It also specifies the format and content for inspection reports to ensure consistency and comprehensiveness, including detailed descriptions of identified issues.

Some approaches address this task from a simpler perspective, focusing solely on identifying hot-spots in PV modules. This can be achieved using Digital Image Processing (DIP) techniques [145], [155] such as adaptive thresholding or advanced Deep Learning (DL) methods [146], [156] like object detection or instance segmentation architectures.

Table 6 summarizes the literature details and results on photovoltaic modules using IRT sensors on UAV platforms, with a focus on crucial details for deep learning model comparison, such as task definition and dataset characteristics.

**TABLE 6. Literature Details and Results on Photovoltaic Modules employing IRT Sensors on UAV Platforms**

| Task Description | Data | | | | Model | | | |
|---|---|---|---|---|---|---|---|---|
| | Sample | Size | Annotations | Source | Algorithm | Architecture | Performance | Ref |
| Hot-Spot (HS) Detection in PV Modules |  | 14,215 frames<br>640 × 480 x 1<br>336 × 256 x 1<br>320 × 240 x 1 | 3,654 Local HS<br>76 Global HS | Available Upon Request [145] | Explicit DIP | Raindrop | F1 = 77.0% | [145] |
| Hot-Spot (HS) and PV Modules Detection |  | 5,600 frames<br>640 × 480 x 1<br>336 × 256 x 1<br>320 × 240 x 1 | 8,070 Local HS<br>125,722 PV modules | Not Available | Supervised DL | S-YOLOv5 | F1 = 96.6% | [146] |
| PV Modules Thermal Pattern Classification |  | 20,000 IR Patches<br>40 x 24 x 1 | 12 Classes of Anomalies | Openly Available [147] | Supervised DL<br>Supervised DL | Ensemble CNN ResNet based<br>Custom CNN 4 Conv2D | F1 = 74,92%<br>F1 = 73.73% | [148]<br>[149] |
| PV Modules Segmentation |  | 862 frames<br>640 x 512 x 1 | 26,612 PV modules | Code and Data Sample Available [150], [151] | Supervised DL + DIP<br>Supervised DL + DIP | Mask R-CNN<br>Mask R-CNN + GPS + SfM | mAP50 = 87.8%<br>mAP50 = 99.3% | [152]<br>[153] |
| PV Modules Thermal Pattern Classification |  | 4.16 million IR Patches<br>640 x 512 x 1 | 12 Classes of Anomalies | Available Upon Request [154] | Supervised DL<br>Semi--Supervised | ResNet-50<br>ResNet-34 + kNN Classifier | F1 = 90.68%<br>AuROC from 73.3% to 96.6% | [152]<br>[154] |

Vincenzo Carletti *et al.* [145] relied on DIP to identify hotspots in PV modules from UAV-collected thermal videos. PV module masks were obtained via Canny edge detection, with bounding boxes refined using clustered, angle-filtered Hough lines. Local hotspots were identified within these grid regions using a raindrop algorithm to find absolute and local minima in inverted pixel values. They proposed a tracking method based on minimum Euclidean distance between bounding box centers across frames. If the distance exceeded a threshold, a new ID was assigned or deleted. To handle intermittent translations, "ghost modules" allowed temporary ID deletion within a frame limit. Hotspot detections were averaged per module across frames to produce a "global detection". The approach was tested on 15 manually labeled videos (available on request). Using IoU > 0.4 as the true positive criterion, they achieved F1-scores of 0.87 for module detection, 0.72 for local hotspots, and 0.77 for global hotspots.

Qiuping Zheng *et al.* [146] created their own annotations for a subset of Carletti *et al.*'s videos [145], labeling 125,722 PV modules and 8,070 hotspots across 5,600 frames. They compared several DL detectors with over 20M parameters to their proposed S-YOLOv5. This lightweight YOLOv5 variant replaced the original feature extractor with the earlier *Focus* structure [157], used *ShuffleNetv2* units in the backbone, and simplified feature fusion in the neck by combining outputs from the 2nd and 4th layers (FPN-style). These changes reduced parameters from over 7M to 3.71M and improved mAP from 0.944 to 0.981. However, no class-wise metrics were reported: of 8,070 hotspots, total FNs and FPs were 3,746 and 3,902, suggesting the high mAP likely reflects strong PV module detection rather than hotspot performance.

This approach has the advantage of straightforward implementation, allowing any video frame to be used as input data for processing, either offline or on an embedded system. However, this simplicity comes with limitations. These methods may not generate valuable information for subsequent maintenance procedures, which require specific details such as the failure location, the impact on power generation, and root cause analysis along with possible repair procedures. A more comprehensive approach considers the system's ability to identify a set of anomaly classes, such as the 12 defined in IEC TS 62446-3 [48], which link thermal patterns to the underlying causes of failure to support diagnosis and repair.

Minhhuy Le *et al.* [148] developed an ensemble DL model for classifying PV thermal patterns using a public dataset [147] of 20,000 IRT cropped images of PV modules, labeled as healthy or as one of 11 anomalies. A validation set used 20% of samples per class, while training data imbalance was addressed via SMOTE oversampling, generating over 83% synthetic data. The authors tested weighted focal loss and on-the-fly augmentation, achieving best results with augmented data and an ensemble of 15 ResNet-based CNNs. The model reached 94.4% accuracy for binary classification (all anomalies combined) and 85.9% for multiclass, with an extrapolated balanced accuracy of 72.75%, that accounts for class prevalence. In another study [149], the authors proposed CNN-PVE, a custom 5-layer CNN with 2.7 million parameters for embedded systems. Quantizing the weights to 16-bit enabled deployment on an NVIDIA Jetson Nano, achieving 954 fps on RaptorMaps dataset [147] images. Accuracy dropped just 0.5% from their prior ensemble CNN [148], with an additional 0.05% after quantization, resulting in 85.35% global and 71.25% balanced accuracy. During qualitative evaluation, they found some ground truth labels misaligned with dataset instructions, raising concerns about annotation noise and its effect on model performance.

Lukas Bommes *et al.* [152] collected 231,172 frames (8 hours) of thermographic video from seven large PV plants using a UAV with GPS. They annotated 862 frames with segmentation masks and bounding boxes for 26,612 PV modules, using 93% for fine-tuning a Mask R-CNN and 7% for validation, achieving an AP of 99.55% at IoU 0.5. A homography was computed by fitting a minimum-perimeter enclosing quadrilateral to each segmentation mask, mapping it to a rectangle. This enabled the creation of 4.3 million IR patches for 107,842 distinct PV modules, which were tracked and grouped by ID. In addition to reducing labeling effort 40-fold, tracking allowed filtering of sun reflections, ensured valid dataset splits, and supported majority voting in thermographic pattern classification, achieving a 90.68% F1-score with ResNet-50. In another study, L. Bommes *et al.* [154] approached fault detection as an unsupervised domain adaptation problem, addressing domain shifts between PV plants under different environmental conditions—often overlooked in prior work. They used labeled data from a source plant to predict on unlabeled data from a target plant. A ResNet-34 trained with supervised contrastive loss generated low-dimensional embeddings, which a k-nearest neighbor classifier used to detect anomalies. This method achieved AUROC scores between 73.3% and 96.6% across source–target combinations without hyperparameter tuning. Identifying another key problem in the integration of fully automated inspections with O&M procedures, Bommes *et al.* [153] developed a tool, illustrated in Fig. 15, for the extraction and georeferencing of PV modules from aerial IR videos. They compensated for the low resolution of the GPS receiver device by interpolating geocoordinates from key frames. Tracking and triangulating each segmented module allowed them to successfully map 99.3% of the PV plant modules.

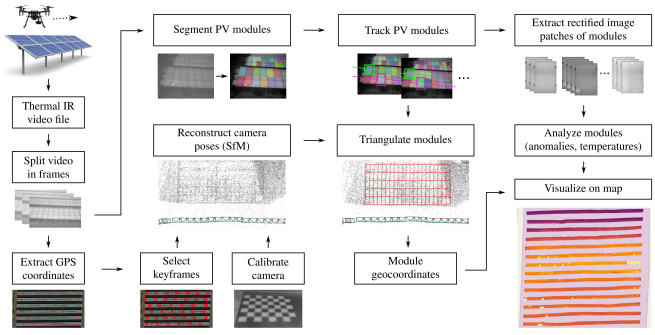



**FIGURE 15. Overview of Bommes method for automatic extraction and georeferencing of PV modules from aerial IR videos, adapted from [153].**

IR imaging detects damaged cells by capturing the IR radiation emitted when solar cells stop converting energy into electricity due to disconnection from the circuit. However, the resolution limitations of IR cameras can hinder the detection of minute defects, such as microcracks that have not yet impacted conversion efficiency. A possible solution involves using electroluminescence (EL) imaging, which offers higher resolution and reveals premature defects. While effective, EL imaging requires trained specialists and manual inspection, making it time-consuming and costly. The potential for automated defect classification in EL images presents an opportunity to overcome these challenges.

EL works by applying a voltage bias to a PV cell, generating an electric field within it. Defects, cracks, or non-uniformities in the silicon can cause localized variations in this field. When carriers (electrons and holes) recombine in these areas, they emit light—a phenomenon known as electroluminescence. Dark regions or patterns in EL images may indicate defects, cracks, or other irregularities. EL imaging is used during manufacturing to identify defects early and ensure the production of high-quality PV cells. In the field, it can help identify early failures in installed PV panels and prevent performance drops.

The IEC TS 60904-13 standard [164] provides guidelines for EL imaging of PV cells and modules to accurately identify and report defects, aiding in PV system performance maintenance. It describes protocols for capturing and processing EL images, calculating signal-to-noise ratios (SNR), and sets minimum SNR values for various applications (outdoor, industrial, laboratory). The standard also includes criteria for classifying and prioritizing anomalies, specifies the format for inspection reports, and provides guidance on classifying potential cracks and interpreting EL images of PV modules.

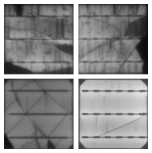
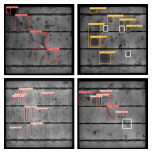
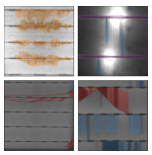
Table 7 summarizes the literature details and results on PV cells using Low Wave Infra-Red (LWIR) sensors on fixed platforms, with a focus on crucial details for DL model comparison, such as task definition and dataset characteristics.

Claudia Buerhop-Lutz *et al.* [172] published an open-source dataset [53] containing 2,624 samples of $300 \times 300$ pixel 8-bit grayscale images of functional and defective PV cells with varying degrees of degradation, extracted from 44 PV modules (26 monocrystalline, 18 polycrystalline). Given the unavailability of actual power degradation measurements, confidence-weighted soft labels were assigned based on expert EL assessments: 1.0 for high-confidence defects, 0.667 for lower-confidence defects, 0.333 for uncertain non-defects, and 0.0 for high-confidence non-defects.

Sergiu Deitsch *et al.*introduced a fully automatic segmentation method [173] for high-resolution EL images of solar cells [53], robust against underexposure and severe defects. This robustness stems from preprocessing techniques and the integration of ridgeness filtering with tensor voting to accurately determine inter-cell borders and busbars. The method achieves 97.80% accuracy, supporting automatic defect classification and power loss prediction. In their follow-up study [158], they compared a hardware-efficient pipeline using hand-crafted features with an SVM, to a more demanding GPU running CNN. Experiments with Dense Sampling and Keypoint Detection showed that KAZE keypoints with VGG descriptors and a linear SVM gave optimal results. The CNN, derived from VGG-19, achieved 88.42% accuracy with 1.125 sec/image CPU inference, while the SVM reached 82.44% accuracy at 0.8125 sec/image.

Recognizing limitations of the ELPV dataset [53], Binyi Su *et al.* [160] introduced PVEL-AD, a dataset of electroluminescence images of photovoltaic cells for anomaly detection. Collected in an industrial setup, it contains 36,543 near-infrared images at $1024 \times 1024$ resolution. The dataset includes 37,380 ground truth bounding boxes obtained using a YOLOv5 model trained on 3,629 annotated images. They inferred unlabeled data with this model and refined the annotations using *LabelImg* [174], reducing annotation workload by two-thirds. Their best anomaly detection results on the most

**TABLE 7. Literature Details and Results on Photovoltaic Cells employing LWIR Sensors on Fixed Platforms**

| Task Description | Data | | | | Model | | | |
|---|---|---|---|---|---|---|---|---|
| | Sample | Size | Annotations | Source | Algorithm | Architecture | Performance | Ref |
| PV Cells Classification |  | 2,624 samples<br>300 x 300 x 1 | 4 Classes of Pseudo Degradation Level | Openly Available [53] | DIP + ML Supervised | KAZE + SVM | F1 = 82.52% | [158] |
| | | | | | Supervised DL | CNN VGG19 | F1 = 88.39% | [158] |
| PV Cells Defects Detection |  | 36,543 samples<br>1024 x 1024 x 1 | 37,380 bounding box<br>4 Classes | Available Upon Request [159] | Supervised DL | YOLOv5l | mAP50 = 81.93% | [160] |
| | | | | | Supervised DL | ELCN-YOLOv7 | mAP50 = 91.93% | [161] |
| PV Cells Defects Segmentation |  | 17,064 samples<br>165 x 165<br>250 x 250<br>400 x 400 | 9 Classes of Anomalies | Model + Data Openly Available [162] | Supervised DL | DeepLabv3 w/ ResNet-50 | F1 = 69.6% | [152] |
| | | | | | Semi-Supervised DL | Deeplabv3+ | F1 = 84.24% | [163] |

frequent classes reached a mAP of 81.93% with YOLOv5l. For the full dataset, including minor classes (star crack, corner, fragment, and scratch), they achieved a mAP of 65.74% with YOLOv5s, acknowledging significant room for improvement and suggesting potential for few-shot learning strategies.

Fu and Cheng [161] used the PVEL-AD dataset [159] to train and test a YOLOv7 model. They focused on the four most represented classes, creating a single annotation box to enclose each crack, arguing that the original annotations, which split long cracks into segments, were unreasonable. They fine-tuned the last layers of a pre-trained ELCN-YOLOv7 model with a small similar dataset before fitting all layers on the target dataset. Their dataset, architecture, and training modifications improved mAP50 by 5.87%, achieving 91.93% and surpassing other works at the time.

Joseph Fioresi *et al.* [175] proposed automating defect detection and localization in EL images of PV cells using semantic segmentation. They compiled 17,064 EL images with pixel-level annotations to present the UCF-EL dataset [162], addressing class imbalance by merging similar classes (reducing nine defective classes to four) and adding 256 realistically simulated defective images generated with a GAN to oversample minority classes. They tested several models and customized the loss function to increase gradients for small-area instances, with DeepLabv3 achieving 95.4% pixel-wise accuracy and 69.6% class-wise F1-score.

Abhishek Jha *et al.* [163] proposed overcoming the high labor of detailed annotations and advancing automatic photovoltaic defect detection in electroluminescence images using semi-supervised learning on semantic segmentation. They used the UCF-EL dataset [162] to train supervised (SL) and semi-supervised (SSL) models based on a student-teacher methodology, relying on a semi-cross-entropy loss to limit noise from low-confidence predictions. The teacher model's inferences were subjected to weak augmentations (e.g., flipping, rotating), while the student was challenged with strong augmentations, including affine transformations, mixup cropping, and color adjustments. Not surprisingly, with only 20% labeled data, SSL-20 easily outperformed SL-20, since it trained with pseudo-labels from the teacher model. When fine-tuned on the full dataset, SSL-100 outperformed SL-100, with F1-score rising from 82.03% to 84.24%, likely due to strong augmentation. However, SL-100 still outperformed SSL-20, indicating that this dataset does not benefit from soft labels in this approach—likely due to low annotation noise. Validation on a combined dataset from five sources confirmed this trend.

### 2) Wind Turbine Blades

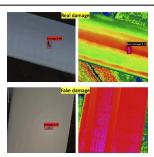
Rotor blades are key components of a wind turbine's power generation system, with their condition significantly affecting performance and accounting for a substantial portion of O&M costs. Among the various NDT methods for inspecting WTB, processing visual data from UAVs has gained prominence due to its accessibility, safety, and economic benefits. Integrating DL for VLI further reinforces this synergy, as more data benefits DL and better models improve data collection.

The Electric Power Research Institute (EPRI) conducted an industry survey to document current practices in categorizing blade damage [176]. The findings shaped a proposed standardized system and highlighted gaps, such as the lack of concrete category definitions. The report defines five damage categories, from minor to severe, with photos, descriptions, system impact summaries, and recommended actions, supporting consistent damage identification and response.

Table 8 summarizes key literature on WTB inspections using RGB or IR sensors on UAVs, focusing on task definitions and dataset characteristics for DL model comparison.

Shihavuddin *et al.* [167] introduced a high-resolution image dataset [165] for WTB surface damage detection, consisting of 558 frames at $5280 \times 2970$ resolution, captured from a single WT tower. Although annotations for multiple damage types, such as erosion and cracks, were kept private, the dataset

**TABLE 8. Literature Details and Results on Wind Turbine Blades employing RGB or IRT Sensors on UAV Platforms**

| Task Description | Data | | | | Model | | | |
|---|---|---|---|---|---|---|---|---|
| | Sample | Size | Annotations | Source | Algorithm | Architecture | Performance | Ref |
| Wind Turbine Surface Defects Detection |  | 558 images<br>5280 x 2970 x 3 | 796 bounding box<br>5 Classes | Openly Available [165], [166] | Supervised DL | Faster R-CNN Incept-ResNet-V2 | mAP50 = 81.10% | [167] |
| | | | | | Supervised DL | YOLOv5s | mAP50 = 83.40% | [168] |
| WTB Surface Defects Detection |  | 1,302 images<br>5456 x 3632 x 3 | 156 bounding box<br>12 Classes | Openly Available [169] | Supervised DL | YOLOv5s | mAP50 = 83.80% | [170] |
| WTB Structural Damage Detection |  | 1,040 images pairs (RGB & IR)<br>6400 x 4000 x 3<br>640 x 512 x 1 | unspecified b.box quantity<br>1 Class | Not Available | Supervised DL | YOLOv7 | mAP50 = 98.50% | [171] |

contributed to advancing WTB automated inspection methods. Using the Faster R-CNN model with Inception-ResNet-v2 backbone, they achieved a mAP of 81.10%. However, the most impactful factor improving model performance was pyramid patching augmentations, which enhanced detection of small and subtle defects across four tested architectures. Despite modest performance, the automated system, when assisting experts, reduced analysis time by 34.5%, demonstrating potential for combining DL with human expertise for more efficient wind turbine inspections.

Imad Gohar *et al.* [168] investigated the impact of improved annotations on defect detection in wind turbine blade images by creating and sharing [166] 796 annotations across five classes: Erosion, Missing Teeth in the Vortex Generator panel, Damage to the Lightning Receptor, Crack, and Paint-Off on a publicly available dataset [165]. Their study highlighted the critical role of accurate annotations in enhancing model performance. They found that multiple annotations of the same defect, rather than a single large annotation, improved detection accuracy in some classes while reducing it in others. Among strategies tested, the D3 labeling approach, dividing the original bounding box into three segments, achieved the highest performance with YOLOv5s, with an mAP@50 of 83.4%. This demonstrates that finer-grained annotations can lead to more effective defect detection in SotA models.

Cong Yang *et al.* [177] proposed a coarse-to-fine image stitching algorithm for efficient WTB inspection. Coarse stitching uses image order, camera distance, and segmentation masks to scale and rotate each image, aligning the bisection angle and matching widths of consecutive masks. Translation, rotation, and scale parameters are then refined, in the fine stitching phase, via regression, minimizing texture and shape losses at overlapping keypoints. They introduced a dataset [169] with drone-captured sequences from 30 blades, totaling 1,302 images at $5456 \times 3632$ pixels resolution. Their approach enables HD defect detection (0.4 mm per pixel) with full coverage of blades up to 80 meters, while avoiding duplicate defect counts. Moreover, the homographically reconstructed images facilitates efficient maintenance planning.

Syed Zeeshan Rizvi *et al.* [170] improved the efficiency of defect detection on WTB by shifting the focus toward dataset pre-processing rather than architectural improvements. This pre-processing step, which removes background using segmentation masks, also enables storage savings. They trained several YOLO models on the original Blade30 dataset [169], as well as on the respective ROIs. Despite potential overfitting in most experiments, indicated by a noticeable discrepancy between training and testing metrics, they achieved an 83.8% mAP@50 on the YOLOv5 Masked dataset test. The use of pre-processing techniques to improve overall model performance is promising, yet requires further investigation.

Weijie Zhou *et al.* [171] addressed false detection rates by implementing a feature fusion module within YOLOv7 that utilizes visible and infrared imagery. They leveraged thermal data to distinguish false damages, like dirty coatings, from real damages such as fissures and cracks, which have distinct thermal patterns due to temperature discontinuities. To work with moderate input sizes and avoid extreme downsampling, they performed global sliding window cropping. They developed a regression network to output a blade bisecting line, avoiding inefficient processing of background-only crops. Validation showed 24.3% mAP improvement with global cropping and an additional 4.9% with regression cropping. Adding the IR fusion module further improved mAP by 5.3%, achieving a final mAP of 98.5%.

### 3) Hydroelectric Dam Walls

Hydroelectric dams are valuable, long-term assets that require regular maintenance to achieve their expected lifespan and maintain operational efficiency. A critical part of this maintenance is the inspection process, which serves as the foundation for identifying potential issues early. VLI inspections, focusing on surface-level observations, are essential for detecting early signs of damage, such as cracks and discoloration. Improving inspection efficiency not only triggers timely maintenance actions but also ensures the dam operates safely, reliably, and in compliance with regulatory standards, extending its life and preventing costly repairs or failures.

Yangtao Li *et al.* [179] propose an automatic classification and localization method for structural cracks in concrete dams, based on deep neural networks and transfer learning. They trained a ResNet50 using 50,476 images of $200 \times 200$ pixels, obtained by cropping, resizing, and combining three structural crack datasets [186]–[188]. They fine-tuned the model using cropped patches from $6000 \times 4000$ pixels images of dam sections 15 and 16, and tested on section 14, achieving a 79.36% F1-score on their patch-level classification model. This sliding window patch classification approach, also followed by ZhiLong Qi *et al.* [189], enabled pseudo-localization of crack defects on the reconstructed image, avoiding the computational effort of object detectors or segmentation models. However, since each patch is treated independently, relationships between patches are not considered, and extraction of higher-level features is not possible. In their subsequent work [181], Yangtao Li *et al.*employed a lightweight semantic segmentation model for underwater crack pixel-wise identification and quantification. They used 14,268 images from four open-access crack datasets to pre-train a model using the LinkNet architecture and fine-tuned it on a private underwater dam crack dataset collected via ROV. This dataset consisted of 855 images at $768 \times 576$ pixels with ground truth masks. They achieved a 92.95% F1-score and 89.24% mIoU at 112 FPS. Their system also quantifies severity of detected cracks, estimating variables such as area, width, and length, calculated based on real physical distance of IR laser measuring points.

Kunlong Hong *et al.* [183] presented a robotic solution to automatically detect and count defect instances, as well as reconstruct the surface of dam spillways, by combining deep learning with a visual 3D reconstruction method. They created a multi-class semantic segmentation dataset [182] of 1,711 augmented images from 450 key frames acquired by an RGB-D sensor on a wall-climbing robot. The dataset includes 6

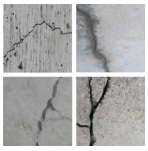
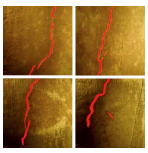
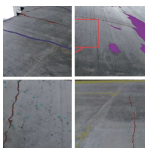
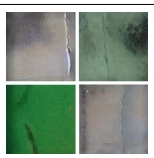
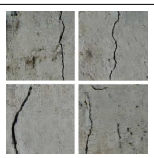
defect classes: cracks, erosion, spots, patched areas, spalling, and power safety cables. The authors used a modified U-Net with pixel-adaptive convolution (PAC) and a conditional random field (CRF) to segment defects of different scales and created a 3D reconstruction model of the inspected asset, using the depth plane for efficient defect instance counting. They acknowledge research on dataset creation methodologies is key for further development of inspection models.

Ben Huang *et al.* [184] proposed an unsupervised image-to-image (I2I) translation method using CycleGAN to generate synthetic underwater images, aiming to expand datasets for training deep learning models. They used 984 above-water images ($640 \times 640$ pixels) and 678 underwater images ($720 \times 576$ pixels) to train several I2I GANs. The best models were selected by evaluating realism and diversity of generated images using SSIM, PSNR, and FID. The usability of generated images was assessed by testing multiple architectures on different tasks using varying proportions of generated images in the training and testing sets. Results showed low performance variance across ratios, leading the authors to claim synthetic underwater images were effective for training. However, by randomly splitting the datasets, it remains unclear whether this approach improves generalization or adds redundant data.

Dong Chen *et al.* [185] enhanced texture details of underwater dam crack images using a multi-feature adversarial loss. They also trained a CycleGAN, but in this case to translate underwater images into above-water images. They utilized 400 unpaired crack images (400 above-water and 400 underwater) from both laboratory and real dam environments. To ensure generated enhanced images preserved structural detail, they separately introduced edge loss and dark-channel loss, matching generated images with corresponding edge and dark-channel feature maps. They tested performance of SOTA models trained on underwater crack images enhanced by their algorithm using an additional 50 non-overlapping underwater crack images. Using YOLOv8s, they achieved an 84.1% mAP, a 58.5% improvement over models trained on original underwater images. As future work, the authors suggest collecting more data, particularly from turbid environments, and annotating other defects, such as spalling and exposed rebars. They also aim to develop lightweight models for mobile devices. However, an aspect not discussed is that, from an information theory perspective, if the GAN's CNNs can encode defect information to generate enhanced versions, it should be possible to detect defects directly from the latent space, without generating enhanced images. This highlights that the true value of the GAN lies in the adversarial learning process itself, rather than in generating enhanced images.

Table 9 summarizes the literature details and results on HDW inspection using RGB on ROV platforms, with a focus on crucial details for DL model comparison, such as task definition and dataset characteristics.

**TABLE 9. Literature Details and Results on Hydroelectric Dam Walls employing RGB Sensors on ROV Platforms**

| Task Description | Data | | | | Model | | | |
|---|---|---|---|---|---|---|---|---|
| | Sample | Size | Annotations | Source | Algorithm | Architecture | Performance | Ref |
| Concrete Dam Structural Cracks Classification |  | 50,476 patches 200 x 200 x 3 + 10,208 patches 200 x 200 x 3 | 2 Classes -Crack -Background | Partially Available [178] | Supervised DL | ResNet50 | F1 = 79.36% | [179] |
| Dam Wall Crack Segmentation |  | 14,268 images 224 x 224 x 3 + 855 images 768 x 576 x 3 | Unspecified Annotated Pixels 1 Classes | Partially Available [180] | Supervised DL | LinkNet | mIoU = 89.24% | [181] |
| Dam Wall Surface Defects Segmentation |  | 450 images 848 x 480 x 4 | 9.7% of Pixels Annotated 6 Classes | Cloud Available [182] | Supervised DL | U-Net | mIoU = 60.3% | [183] |
| Dam Wall Underwater Crack Images Generation and Detection |  | 984 images 640 x 640 x 3 + 678 images 720 x 576 x 3 | Unspecified B.Box Num. 2 Classes | Not Available | Supervised DL | YOLOx | mAP = 84.33% | [184] |
| Dam Wall Above-Water Crack Images Generation and Detection |  | 450 images 256 x 256 x 3 | Unspecified B.Box Num. 1 Class | Available Upon Request [185] | Supervised DL | YOLOv8s | mIoU = 84.1% | [185] |

### B. TRANSPORT AND DISTRIBUTION SYSTEMS

Electrical assets in transport and distribution (T&D) systems, such as power line conductors and structures, high-voltage insulators, and other components, as well as substations and transformers, share common characteristics from the perspective of automated visual inspections. These T&D assets often exhibit high variability in size, shape, texture, and position. The range of defect types is extensive, as is the variation in dimensions of each asset within the system, which consequently affects the optimal acquisition distance for each image.

Common combinations used for data collection in T&D system inspections range from manned to remote-controlled platforms, including ground-based, aerial, and wire-suspended types, all equipped with sensors such as RGB cameras, LiDAR, IR thermography, and UV imaging sensors. These combinations operate at varying capture distances and speeds depending on specific inspection tasks, which may be routine, emergency, comprehensive, or intrusive, as defined in the IEEE 1808 standard [12]. This standard provides guidance focused on the efficient collection and management of inspection and maintenance data for T&D assets. It assists utilities and contractors in developing computer-based systems for managing inspection data, ensuring accuracy, compatibility with GIS systems, and integration with other enterprise software. The standard also emphasizes the importance of risk assessment, data analysis, and the application of AI and ML to improve the detection and management of anomalies.

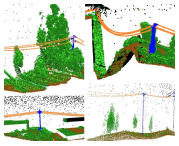
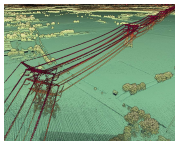
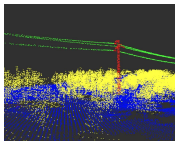
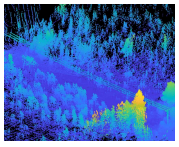
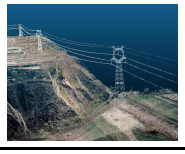
#### 1) Power Lines Conductors and Structures

Automated inspection of power line conductors and structures is vital for system reliability and safety. Integrating UAVs with LiDAR improves monitoring of vegetation, conductor sagging, and tower tilting while reducing manual inspection risks. Table 10 summarizes the literature details and results on PL conductors and structures using LiDAR sensors on mobile platforms, with a focus on crucial details for DL model comparison, such as task definition and dataset characteristics.

Diogo Lavado *et al.* [62] introduced a 3D point cloud dataset collected using a UAV equipped with a LiDAR sensor, covering more than 40,000 km of electrical transmission systems in rural terrain. The dataset comprises 2.595 billion points and 36,000 bounding boxes across 22 classes. The five main classes (Power Line, Suport Tower, Ground, Road, and Vegetation) constitute over 88.5% of the points, with only 1.43% corresponding to electrical assets. In their following work [191], the authors improved Scene-Net [200] by adding three more GENEO [2] kernels and incorporating a multi-class classifier in the model output. Using a white-box feature extractor with just 540 parameters, they not only addressed the explainability problem but also set a new benchmark for parameter efficiency. Their Scene-Net V2 achieved a 50.21% mIoU with 0.26 million parameters, while the State-of-the-Art (SotA) Point Transformer V3 [192] obtained 65.58% mIoU with 46 million parameters.

[2]Group Equivariant Non-Expansive Operators [201]

**TABLE 10. Literature Details and Results on Power Lines Conductors and Structures employing LiDAR Sensors on UAV Platforms**

| Task Description | Data | | | | Model | | | |
|---|---|---|---|---|---|---|---|---|
| | Sample | Size | Annotations | Source | Algorithm | Architecture | Performance | Ref |
| 3D points Classification of Power Lines |  | 2,595 M 3D points along 40,000 km | 5 (22) Classes<br>36,000 B. Boxes | Code + Data Openly Available [190] | Explainable ML | SCENE-Net V2 | mIoU = 50.21% | [191] |
| | | | | | Supervised DL | Point Transformer V3 [192] | mIoU = 65.58% | [191] |
| 3D points Classification of Power Lines |  | 27.4 M 3D points Unspecified Line Length | 5 Classes<br>Unspecified B. Box Num. | Annotations Not Available<br>Data Openly Available [52] | Supervised DL | PSFE-Net | mIoU = 89.90% | [193] |
| | | | | | Supervised DL | SCD-Net [194] | mIoU = 88.63% | [193] |
| 3D points Classification of Power Lines |  | Unspecified 3D points Unspecified Line Length | 4 Classes<br>Unspecified B. Box Num. | Partially Available [195] | Unsupervised ML | Reflectivity Threshold + Cluster PCA | F1 = 95.09% | [196] |
| | | | | | Supervised DL | PointNet++ [197] | F1 = 83.10% | [196] |
| Power Lines Conductors Extraction and Modelling |  | Unspecified 3D points Unspecified Line Length | Not Existent | Not Available | Unsupervised ML | PCA + K-D tree + DBSCAN | Not Evaluated | [198] |
| Power Lines Conductors Distances Assessment |  | Unspecified 3D points 160 km Line Length | Not Existent | Not Available | Unspecified | Unspecified | Not Evaluated | [199] |

Junhua Dong *et al.* [193] introduced PSFE-Net, a semantic segmentation neural network for aerial LiDAR power line inspection. They annotated five classes (ground, pylons, power lines, vegetation, and houses) in a subset of the AHN dataset [52], which contains a transmission power line with 27.4 million points. The proposed network includes a module for symmetry feature extraction designed to enhance pylon classification. PSFE-Net achieved a slight improvement over the previous state-of-the-art, SCD-Net [194], increasing the mIoU from 88.63% to 89.90% on the annotated dataset.

V. Valseca *et al.* [196] presented a real-time LiDAR-based semantic classification method for powerline inspection. They observed that the measured reflectivity of objects is influenced more by their geometry than by their material's inherent reflectivity properties. Using a multi-condition threshold on the eigenvalues resulting from Principal Component Analysis (PCA), they segmented PL conductors and towers from the set of metallic points, and vegetation and soil from the set of organic points. The proposed method was evaluated using DALES dataset [195], as well as their own data collected from more than 30 experiments, with flight distances ranging from 300 to 1,300 meters at 3.5m above the PL. Their method achieved superior inference time and metrics performance compared to PointNet++ [197], obtaining F1-scores of 96.36% and 85.82% for power lines and towers, respectively.

Alexander Kyuroson *et al.* [198] explored autonomous point cloud segmentation techniques for power line inspection, presenting a multi-step approach. In the preprocessing phase, they employed histogram thresholding to filter out ground points and low-elevation outliers. Power lines were subsequently extracted using Principal Component Analysis (PCA), where the eigenvalues of the covariance matrix were used to detect the linear structures of the clusters. To model the segmented power lines, the authors applied polynomial regression, offering a simplified approximation of the catenary curve, especially suitable for small sag angles. While the study presented interesting explanations, it lacks critical details regarding the dataset, testing protocols, and performance metrics. The only reported result is the method's inference time, achieving $0.21\ \mathrm{ms/m^2}$, which marks a notable improvement over the $0.87\ \mathrm{ms/m^2}$ achieved by PointNet++ [197].

Shuo Wang *et al.* [199] used a UAV to acquire 3D LiDAR data for identifying safety hazards across five transmission PL spanning 160 km in the mountainous region of Sanmenxia City, China, identifying 77 vegetation-related hazards. Processing the 3D point cloud data consisted of four steps: denoising, line repair, thinning, and classification. Denoising removed isolated points, and line repair involved fitting a catenary equation to existing points and adding new ones to increase point density. Thinning reduced density in redundant areas, while classification sorted points into four classes. Their experience focused on calculating minimum distances between objects, respecting local safety standards to identify hazards at three severity levels. They identified and reported to the local power department, a total of 54 general hazards, 22 major hazards, and one emergency hazard.

LiPowerline™ software, developed by GreenValley International® [202], offers advanced tools for PL analysis, including automatic classification of PCs with over 95% accuracy for transmission lines and 85% for distribution lines. In addition to post-processing for correcting georeferencing data and completing missing points for more consistent visualizations, it provides over 15 measurement functions, including minimum distance, sag analysis, tower inclination, and cross-arm height difference.

While LiDAR provides accurate spatial measurements of structures, managing 3D point clouds poses challenges due to large data volumes, complex processing, and the need for skilled interpretation. In contrast, other studies use RGB images to detect power line encroachments, focusing on simpler tasks like identifying foreign objects or bird nests, which trade some precision for easier processing.

Michele Gazzea *et al.* [77] propose a hybrid approach for automated monitoring of vegetation encroachment along power lines by combining high-resolution satellite imagery with LiDAR projections as ground truth. This method leverages the frequent coverage and cost-effectiveness of satellite imagery alongside the detailed accuracy of LiDAR for model evaluation. Conducted along a 22 kV sub-transmission line spanning 26 km in western Norway, the study used 0.5-meter resolution satellite images from 2017 and 2018, cropped into $80 \times 80$ pixel windows centered on the georeferenced power line corridor, while LiDAR data from 2019 generated 2D projections of the 3D point cloud, including only vegetation to produce a binary mask. The model includes two blocks: a Feature Extractor (FE) and Multi-Layer Perceptron (MLP) for pixel-wise semantic classification, and a CNN outputting superpixel clusters trained with manual labels. A Tree-Density Index, based on tree distance from power lines, was computed for both satellite and LiDAR data, achieving weighted average F1-scores of 69.86% and 74.48% for Pleiades-1 and WorldView-2 images, respectively.

Using satellite images for vegetation monitoring offers wide coverage, cost-effectiveness, and temporal analysis but faces challenges such as resolution limits, cloud interference, processing complexity, and latency in data acquisition.

Mateo Cano-Solis *et al.*created the VEPL dataset [204] for semantic segmentation of vegetation encroachment in power line corridors. RGB images were captured by UAVs flying at 30 m along 2.4 km of low-voltage distribution lines, yielding 532 orthomosaic images of $256 \times 256$ pixels, augmented geometrically and spectrally. They developed VEPL-Net [209], a deep learning ensemble for automatic segmentation using UAV imagery. Testing U-Net and DeepLab architectures with BCE and Tversky losses, the best model—DeepLab with Tversky loss on the augmented dataset—achieved 68.25% mean Intersection over Union (mIoU).

The use of monocular images, for proximity detection tasks, implies some assumption about safety distances allowed, such as the vertical clearance matching the PL height, which is not always the case. To address this limitations, other forms of data capture might be considered, beside LiDAR.

Shuaiang Rong *et al.* [210] employed stereovision techniques for intelligent detection of vegetation encroachment around PLs. Their framework integrates binocular vision sensors mounted on towers with embedded image processing for real-time monitoring. PLs were detected using the Hough transform, while vegetation—specifically trees was identified by a Faster R-CNN model trained on 70 tree images and tested on 20, achieving 95% accuracy. Sensors, remotely operable with zoom and tilt, are calibrated using known PL dimensions and positions, optimizing a transformation matrix to minimize error between real-world and reconstructed feature points. Vegetation key points extracted by SIFT enable calculating the distance to PLs with under 5% error.

Despite the limited availability of usability and impact metrics in real-world industrial scenarios, another common application of automated inspection in power line maintenance, as highlighted in the literature, is the detection of foreign objects such as bird nests, plastic bags, and balloons that become entangled in exposed conductors. Some studies [211], also include categories like fire, people, and monkeys.

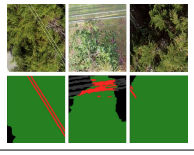
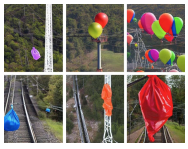
Zhichao Chen *et al.* [207] created a dataset for foreign object detection on transmission lines by combining three synthetic data sources. The first source manually adds foreign objects to public railroad images; the second uses a text-to-image generative model; the third composites backgrounds, objects, and masks with color harmonization. The dataset includes 14,615 images and 40,541 annotations. After testing several SOTA models, YOLOv8-large achieved the highest mAP@50 at 94.2%. Deployment on real images showed satisfactory results. However, the study did not address the system's operational impact, since detecting target objects that do not interfere with power lines could cause false alarms.

Several studies [212]–[215] build on this scenario to improve object detector architectures and algorithms, enhancing feature recognition. However, models risk overfitting to training data, and generalization assessment remains difficult due to dataset similarities. Practical value declines if focus shifts to metric optimization over real-world utility. Achieving relevance requires thoughtful task framing and class selection to integrate detection into maintenance workflows.

Moheddin Sumagayan *et al.* [208] emphasize an integrated system addressing both image acquisition and processing. Their work includes visual power line tracking for semi-autonomous navigation, asset identification for targeted inspection, and foreign object detection to trigger interventions. Efficient data acquisition is supported by offline route planning and real-time detection for trajectory adjustments and collision avoidance. Keypoints along power lines were detected using a modified Point Instance Network (PINet), while utility assets and foreign objects used YOLOv5. They generated three datasets from video frames: 595 images for power lines, 1,040 images for six utility asset classes from 12 locations, and 1,505 images for three foreign object classes collected at a test site. Each image was augmented ninefold using geometric, color, and filter transformations, expanding the dataset tenfold. The augmented sets were split 8:1:1 for training, validation, and testing. Results include a 72% F1-score for power line detection, mAP@50 of 96.8% for utility assets with YOLOv5-m (640×640×3), and mAP@50 of 86.39% for foreign objects with YOLOv5-l (1280×1280×3).

Table 11 summarizes the literature details and results on PL conductors and structures using RGB sensors on different platforms, with a focus on crucial details for DL model comparison, such as task definition and dataset characteristics.

**TABLE 11. Literature Details and Results on Power Lines employing RGB Sensors on UAV Platforms**

| Task Description | Data | | | | Model | | | |
|---|---|---|---|---|---|---|---|---|
| | Sample | Size | Annotations | Source | Algorithm | Architecture | Performance | Ref |
| Segmentation of Vegetation near Power Lines |  | 5 Images<br>800 x 800 x 3<br>+ 10 Images<br>400 x 400 x 3<br>26 km PL Length | 1 Class<br>Unspecified Pixels Num. | Not Available | Supervised ML + DL | FE + MLP + Custom CNN | F1 = 72.17% | [77] |
| Segmentation of Vegetation and Power Lines |  | 532 Images<br>256 x 256 x 3<br>2.4 km PL Length | 2 Classes<br>Unspecified Pixels Num. | Openly Available [203] | Supervised DL | Unet +BCE | mIoU = 66.35% | [204] |
| | | | | | Supervised DL | DeepLab + Tversky | mIoU = 68.25% | [204] |
| Foreign Object Detection near Power Lines |  | 14,615 images from 1920 x 1080 x 3 to 2048 x 2048 x 3 | 4 Classes<br>40,541 B. Box Num. | Code + Data Openly Available [205], [206] | Supervised DL | YOLOv8-l | mAP50 = 94.2% | [207] |
| | | | | | Supervised DL | YOLOv6-s | mAP50 = 93.9% | [207] |
| Foreign Object Detection Obstructing Utility Assets |  | 3,140 Original Images Unspecified Resolution | 9 Classes<br>Unspecified B. Box Num. | Not Available | Supervised DL | YOLOv5-l 1280 input | mAP50 = 86.39% | [208] |
| | | | | | Supervised DL | YOLOv5-x 640 input | mAP50 = 85.43% | [208] |

### 2) High-Voltage Insulators and Other Components

The integrity of High Voltage (HV) insulators and other components is crucial for uninterrupted service in electrical networks. UAVs equipped with RGB cameras provide a flexible solution for inspecting these assets, allowing for detailed visual assessments from various angles and heights. This technology aids in the early detection of wear and potential failures, contributing to more effective maintenance strategies.

Table 12 summarizes the following literature results of DL on HV Insulators using RGB sensors on UAV platforms.

The Electric Power Research Institute (EPRI) [216] is fostering collaboration to advance utility infrastructure inspection by promoting software development competitions and sharing labeled datasets on major data platforms, including IEEE Dataport [217]. Despite no public works being found using the Electric Transmission Imagery [218] and only one study [219] using the Distribution Inspection Imagery [220], the Insulator Defect Image Dataset (IDID) [47] has been cited in over 40 works according to a Google Scholar search conducted at the time of this review. Its high resolution and close-up views of real operational environments make it one of the best publicly available datasets for HV insulators.

Since the IDID-v1.2 [47] was shared in a competition context, the test set annotations were kept private. Consequently, the available test set contains 88 unlabeled images, while the training dataset includes 1,600 annotated images across 4 classes: String Insulator, Good Shell, Broken Shell, and Flashover Shell. However, only 400 images are unique, with the remaining images generated through offline mirroring augmentation, which may require additional care in how the validation of the trained models is performed. These dataset characteristics and others were properly identified in the study conducted by Fernando Lopes *et al.* [221], leading them to proper careful selection of the images in the validation set. They fine-tuned two SOTA models using 3 subclasses (Good, Broken, and Flashover), achieving validation mAP50 scores of 87.9% and 87.2% with YOLOv8s and Faster R-CNN X101-FPN, respectively. To address the absence of test annotations and quantitative metrics, the authors conducted a qualitative analysis of the prediction results, providing a detailed description of the models' generalization capabilities.

Jianfeng Zheng *et al.* [223] utilized the images from IDID [47] but created a new set of annotations [222] using different implicit rules. Instead of using bounding boxes enclosing the entire asset, they annotated defects present in individual insulator discs, which led to discarding the Good Disc class. They compared the performance of various SOTA object detectors against their proposed improved YOLOv7 model. The ablation study revealed performance improvements for every component, achieving a mAP50 of 93.8%. However, if the validation split was randomly defined, predictions would be made on mirrored images of those used in training. In practice, such metrics may indicate overfitting rather than true generalization. This random dataset split was explicitly applied to the IFDD dataset [222] in the work presented by Dong Li *et al.* [238], which was later used by Xiao Wang *et al.* [239].

Xian Tao *et al.* [240] addressed the scarcity of defect images in real inspection environments by synthesizing defective insulator images to create the *Chinese Power Line Insulator Dataset (CPLID)* [224]. The dataset contains 848 RGB images with a resolution of $1152 \times 864$ pixels, including 600 original images of functional composite string insulators. To generate the defective set, a segmented defective insulator with missing discs was pasted onto various backgrounds, resulting in 248 synthetic images. While this dataset provides a valuable resource for training models to detect synthetic objects in real backgrounds, the relatively limited diversity of defects may limit its ability to fully assess generalization to real-world defect scenarios. In 2018, they [240] proposed a cascade architecture: first detecting the composite insulator string from the background, and then detecting the missing disc within the insulator string. Using ResNet-101 as the backbone in an undisclosed detection architecture, the study reported F1-scores of 93.3 and 90.9 for the two steps, respectively. However, given the cascade nature of the model, the absence of a combined performance metric makes it difficult to fully assess the overall effectiveness. A simple approximation based on the product of the two reported scores ($0.933 \times 0.909 \approx 0.848$) suggests a more modest end-to-end performance.

Bruno Souza *et al.* [241] used the IDID [47] to develop a Hybrid-YOLO model combining object detection for insulator discs with a CNN classifier to distinguish defective from functional assets. They ran the experiment of training the YOLOv5-x model for 780 epochs, 50 times, with random weight initialization, obtaining an average mAP50 of 99.19% with insignificant variance on the validation set. On the test set, the F1-score dropped from 98.84% to 95.5%. Using the original dataset annotations, they performed image cropping to produce a classification dataset consisting of 12,827 cutout images. Among the several CNN architectures tested for the classifier, ResNet-18 achieved the best F1-score of 96.22%. The reported F1-score for YOLOv5x across all classes was 94.49%, which the authors considered inferior to their classifier-based cascade. However, it appears that the evaluation did not consider the combined effect of the cascade's sequential steps; a simple multiplication of the individual test results ($0.955 \times 0.9622 \approx 0.919$) would indicate a lower F1-score than YOLOv5x. Moreover, the low standard deviation in the validation results supports the tested hypotheses. However, the hypotheses under test might not represent the model's generalizability as accurately as testing on different images would.

Zhiyong Tao *et al.* [225] presented even superior results, using the CPLID [224] and IDID [47] datasets. They introduced SnakeNet, a modification of the YOLOv8n model, replacing several architectural modules, including the DSConv-Neck, C2MA, and GCVNet modules, as well as the loss function. This reduced the model's size from 3.2M to 2.9M parameters and improved mAP50 performance from 97.5% to 99.2% on IDID [47] and from 99.4% to 99.5% on CPLID [224]. An interesting direction for evaluating generalization would have been to train on one dataset and test on the other. Instead,

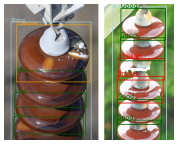
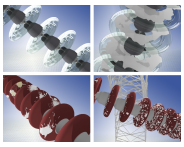
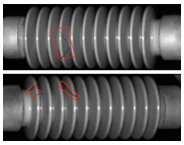
the evaluation was performed using test sets that appear to include augmented variants of the training data. This setup introduces potential data leakage, which may lead to overestimated performance and does not provide a clear measure of cross-dataset generalization.

Reinaldo Bianchi *et al.* [242] created a synthetic HV insulators dataset [243] using a game development engine and publicly available CAD models of the most common insulator types. By manipulating virtual cameras, they were able to obtain over 60,000 images with a resolution of 512x512, along with the automatic generation of ground truth masks. The dataset is divided into two sets: The Image Segmentation Set, which includes 47,286 images categorized by insulator material (ceramic, polymeric, and glass) and landscape type (mountains, forests, plantations, rivers, cities, and deserts), and the Image Classification Set, which contains 14,424 images simulating common insulator string contaminants such as salt, soot, bird excrement, as well clean insulators.

Ebru Ergun *et al.* [228] evaluated the cleanliness of HV insulators using the classification set from the synthetic dataset [243]. Their best-performing hybrid model combined VGG16 and GoogleNet as feature extractors with a Random Forest (RF) classifier, achieving a balanced F1-score of 98.54%, as calculated from the presented confusion matrix, on a 5-fold cross-validation. However, the absence of testing on other datasets limits the evaluation of the model's generalization capabilities in real-world contexts.

Hericles Ferraz *et al.* [227] extended their team's previous work [242] by training and testing a DL model for the automated classification of electrical insulator cleanliness. For the segmentation task, the top-performing architecture was a LinkNet with a VGG16 backbone, achieving a Dice coefficient of 95%, equivalent to a mean Intersection over Union (mIoU) of 90.48%. For the classification of dirty defects, the best test results were obtained with a custom CNN, achieving a balanced F1-score of 97.32%. They performed validation and

**TABLE 12. Literature Details and Results on on High Voltage (HV) Insulators employing RGB Sensors on UAV Platforms**

| Task Description | Data | | | | Model | | | |
|---|---|---|---|---|---|---|---|---|
| | Sample | Size | Annotations | Source | Algorithm | Architecture | Performance | Ref |
| Functional and Defective Discs Detection in HV Insulators Strings |  | 1688 Images from 1618 x 1080 x 3 to 7360 x 4912 x 3 | 4 Classes<br>17,116 B. Box Num. | Open Available [47] | Supervised DL | Faster R-CNN | mAP50 = 87.2% | [221] |
| | | | | | Supervised DL | YOLOv8s | mAP50 = 87.9% | [221] |
| Defective Fraction Detection in HV Insulators Discs |  | 1600 Images from 1618 x 1080 x 3 to 7360 x 4912 x 3 | 3 Classes<br>5,373 B. Box Num. | Open Available [222] | Supervised DL | YOLOv7 | mAP50 = 90.1% | [223] |
| | | | | | Supervised DL | Improved YOLOv7 | mAP50 = 93.8% | [223] |
| HV Insulators Strings and Missing Discs Detection |  | 848 Images 1152 x 864 x 3 | 2 Classes<br>1,325 B. Box Num. | Open Available [224] | Supervised DL | YOLOv8n | mAP50 = 99.4% | [225] |
| | | | | | Supervised DL | SnakeNet | mAP50 = 99.5% | [225] |
| HV Insulators Dirty Defect Classification |  | 47,286 Images 512 x 512 x 3 + 14,424 Images 512 x 512 x 3 | 4 Classes of Dirty HV Insulators | Open Available [226] | Supervised DL | Custom CNN | mIoU = 97.32% | [227] |
| | | | | | Supervised DL | VGG16 + RF | mAP50 = 98.54% | [228] |
| Catenary Insulators Defective Pixels Classification |  | 4,293 Images 512 x 512 x 1 | 7 Classes<br>10% Defects | Data + Code Open Available [229] | "Unsupervised" DL | FastFlow [230] | F1 = 63.07% | [231] |
| | | | | | "Unsupervised" DL | MAE [232] | F1 = 68.89% | [231] |
| MV Insulators Broken Defect Detection |  | 1,236 Images 512 x 512 x 3 | 1 Class<br>2,438 B. Box Num. | Open Available [233] | Supervised DL | YOLOv7 | mAP50 = 78.5% | [234] |
| | | | | | Supervised DL | ID-YOLOv7 | mAP50 = 85.7% | [234] |
| Object Detection HV Insulators |  | 1100 Images 640 x 640 x 3 | 1 Class<br>Unspecified B. Box Num. | Data + Code Open Available [235], [236] | Supervised DL | YOLOv8m | mAP50 = 92.0% | [237] |
| | | | | | Supervised DL | YOLOv8-Gold | mAP50 = 94.1% | [237] |

test splits prior to augmentation, which was applied on-the-fly exclusively to the training set. However, the strong similarity between images limited the assessment of the model's generalization capabilities. To address this limitation, they conducted a final test using 50 real images from diverse sources, achieving an mIoU of 84.61% for insulator disc segmentation. Additionally, they presented qualitative results on real images containing various dirt-related defects, further demonstrating the model's performance.

Tao Zhang *et al.* [231] collected catenary insulator images captured on real railway lines to construct the Catenary Insulator Defect (CID) dataset [229]. They trained an unsupervised Masked Auto-Encoder (MAE) with 3,900 defect-free samples and tested it with 160 real defects and 233 simulated defects, distributed across six defect classes: breakage, contamination, cracks, dirt, missing components, and sheltering. The MAE model was intentionally overfitted to defect-free images during training and used to reconstruct input images after a partial patch occlusion. During inference, defective images were reconstructed as normal assets, and the Structural Similarity Index Measure (SSIM) was calculated between input and output pixels to determine an anomaly score and detect defects. They reported an F1-score of 99% for image classification and 68% for image segmentation on the combined test set. Despite these methods being often described as unsupervised learning, the data used is annotated, even though it belongs to a single "defect-free" class. Acknowledging this, they further merged all existent classes to conduct an experiment they termed "completely unsupervised". To some degree, this still constitutes a form of a supervised task, as the model is ultimately trained, in this case with 10% noisy data, to output a generated image of the defect-free class.

Bojian Chen *et al.* [234] redesigned the YOLOv7 architecture to improve the detection of broken insulators in power distribution networks. To validate their modifications, which included data augmentation, network structure, and loss function adjustments, they used the SU22kV dataset [233] after addressing inaccuracies in the annotated data. They performed ablation experiments, achieving improvements across all proposed modules, culminating in an mAP50 of 85.7%. To further evaluate the effectiveness of their proposed ID-YOLOv7 model for object detection tasks, they trained it on the PASCAL VOC 2007 dataset, achieving an impressive result of 90.3% mAP50 and establishing a new SotA benchmark.

Benelmostafa *et al.* [237] proposed improvements to the YOLOv8 model for detecting broken glass insulators and constructed the VPMBGI dataset [235] by merging other publicly available datasets and carefully annotating 1,010 images. Their modification incorporates the Gather and Distribute mechanism for fusing different feature maps within the Neck component of YOLOv8. The validation results demonstrated an improvement of 2.1% in mAP50, reaching 94.1%. However, the existence of multiple images of the same asset with low rotation and translation variations across the training and validation splits, which is often neglected in favor of achieving higher quantitative metrics, might not represent true generalization ability. Despite the model weights no longer being available, the authors shared the model and the code used for training [236], allowing further comprehensive testing.

Acknowledging the lack of publicly available real-world datasets for power line components, André Silva *et al.* [245] introduced the Power Line Asset Inspection Dataset, InsPLAD [244]. This dataset comprises 10,607 full-HD images annotated into 17 asset classes, resulting in 28,933 annotations for object detection. Additionally, they created two derived datasets of augmented cropped patches: one containing only functional assets for unsupervised anomaly detection, and another for image classification, with defect labels for five assets, including corrosion, broken components, and bird nest presence. They benchmarked several models, where the best results achieved include 89.1% mAP50 using RetinaNet for detection and 95.4% average balanced accuracy using EfficientNet for classification. In the following study [246], the authors explored attention modules for image-level anomaly detection using, among other publicly available datasets, a subset of their own dataset [244], containing 11,662 images, of which 402 were defective samples used only for testing. They introduced AttentDifferNet by improving DifferNet and achieved their best result of 94.34% AUC with the Squeeze-and-Excitation Networks (SENet) module. However, this metric does not provide a comprehensive overview due to the severe class imbalance in the test set (e.g., yoke suspension: 96% good / 4% defective).

Chuanyang Liu *et al.* [247] state that rust defects on power components pose significant hidden dangers to the safe operation of transmission lines. To address this, they modified the YOLOX model architecture to detect rust on dampers and clamps. Their modifications included the integration of the Coordinate Channel Attention and Residual Module (CCARM), the Receptive Field Block (RFB), the Efficient Convolutional Block Attention Module (ECBAM), and a focal loss function. While the standard YOLOv7 achieved a mAP of 91.65% on the test set, their RD-YOLO model attained a slightly higher mAP of 92.33%.

Jun Wu *et al.* [248] modified a YOLOv5 model to detect corrosion in transmission line components, integrating several architectural components (WA-PANet, PSA, RFENet, and EIoU Loss). Frames were extracted from a single video, resulting in 2,705 images with a resolution of $5184 \times 3888$ pixels. These were annotated into four object classes: anti-vibration dampers and insulator fittings, each categorized with and without corrosion. Despite stating that shuffling the dataset before splitting it into training, validation, and test sets helped improve model generalization, it would be important to specify how many distinct assets were covered in the video.

Anicetus Odo *et al.* [249] investigated the effects of occlusion, scale, and multi-task detection in asset management of power lines (PL). The proposed system utilizes detected instances in tower images to create resized patches for a tower classification model, which binarizes a 5-level fault index criteria rating (CR) (1 to 3 = healthy, and 4 to 5 = unhealthy). The original dataset, provided by Northern Powergrid, derives

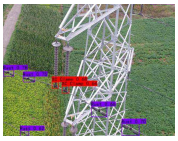
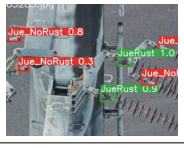
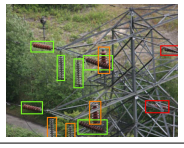
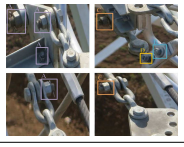
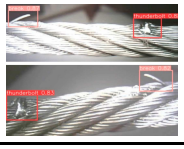
from helicopter inspections of three power lines, consisting of $5616 \times 3744$ pixels images from eight different views of 7,219 PL tower assets. A subset of the data was used, including 363,859 annotated objects across eight PL component classes. From this subset, 1,600 images from 200 towers were used to train a Mask R-CNN detector. The test set included 949 clear insulators, 115 partially occluded insulators, and 98 heavily occluded insulators. The AP50 for clear assets was 88.7%, while the mAP50 dropped to 83.8% for occluded assets. High-risk tower comments in inspection reports were used to identify a subset of closer-plane images, where insulators were detected with 96.7% AP50. For the tower classification system, 25,000 images from 1,000 towers were selected, with 200 images from each of the five CR conditions. The balanced F1-score achieved with ResNet50v2 was 84.13%, while EfficientNetB0 yielded 86.51%, using only the "Insulator" class for tower condition classification.

Motivated by the challenge of detecting ultrasmall defects in bolts during transmission line inspections, Peng Luo *et al.* [84] developed a method to fully leverage the high-resolution images commonly captured in modern inspections. The dataset comprises 1,852 images with 14,635 annotated objects across six classes, with relative object areas ranging from 0.1% to 0.01%. The Ultrasmall Object Perception Module (UOPM) processes the downsampled input to identify salient regions, which are mapped back to the original image to generate high-resolution local images. These are subsequently processed by the Local Bolt Detection Module (LBDM) at the original resolution. Heatmap visualizations demonstrated that the Self-Attention Semantic Feature Extraction Branch (SASFEB) enhanced feature map extraction in the convolutional layers. Despite requiring longer inference times, their Ultrasmall Bolt Defect Detection Model (UBDDM) outperformed other SotA architectures, achieving a 78.6% mAP compared to 47.2% with Faster R-CNN.

Yuting Wu *et al.* [251] studied another application of automated inspections for electrical grid assets by training a modified YOLOv8 model to detect damages in power line conductor cable wiring. They combined a public dataset [250], containing 1,318 images $1080 \times 1080$ pixels resolution with 1,512 annotated objects across two classes (break and thunderbolt), with private data and further expanded it through augmentation. They achieved a 91.1% mAP50 using their modified, unspecified-size variant of YOLOv8, compared to 90.1% obtained by the EfficientDet model.

Table 13 summarizes the literature details and results on several PL components using RGB sensors on UAV platforms, with a focus on crucial details for DL model comparison.

**TABLE 13. Literature Details and Results on several Power Lines Components employing RGB Sensors on different Platforms**

| Task Description | Data | | | | Model | | | |
|---|---|---|---|---|---|---|---|---|
| | Sample | Size | Annotations | Source | Algorithm | Architecture | Performance | Ref |
| Defective Classification of PL Components |  | 10,607 Images<br>1920 x 1080 x 3<br><br>39,150 patches | 17 Classes<br>28,933<br>B. Box Num.<br>5 Classes<br>w/ Defects | Open Available [244] | "Unsupervised" DL | DifferNet | AUC = 90.5% | [245] |
| | | | | | "Unsupervised" DL | AttentDifferNet | AUC = 94.34% | [246] |
| Rust Detection on PL Dampers and Clamps |  | 3400 Images<br>416 x 416 x 3 | 4 Classes<br><br>Unspecified<br>B. Box Num. | Not Available | Supervised DL | YOLOv7 | mAP = 91.65% | [247] |
| | | | | | Supervised DL | RD-YOLO | mAP = 92.33% | [247] |
| Rust Detection on PL Dampers and Fittings |  | 2705 Images<br>5184 x 3888 x 3 | 4 Classes<br><br>Unspecified<br>B. Box Num. | Not Available | Supervised DL | YOLOv5 | mAP = 90.15% | [248] |
| | | | | | Supervised DL | PWR-YOLOv5 | mAP = 95.37% | [248] |
| PL Tower Health Condition Classification |  | Unspecified<br>Images Num.<br>5616 x 3744 x 3<br>386,401 patches<br>Unspecified Resol. | 2 Classes<br><br>Healthy<br>Unhealthy | Not Available | Supervised DL | ResNet50v2 | F1 = 84.13% | [249] |
| | | | | | Supervised DL | EfficientNetB0 | F1 = 86.51% | [249] |
| Bolt Defect Detection on PL Fittings |  | 1,852 Images<br>Unspecified Resol. | 6 Classes<br><br>14,635<br>B. Box Num. | Not Available | Supervised DL | Faster R-CNN | mAP = 47.2%% | [84] |
| | | | | | Supervised DL | UBDDM | mAP = 78.6% | [84] |
| PL Conductors Damage Detection |  | 1,318 Images<br>1080 x 1080 x 3 | 2 Classes<br><br>1,512<br>B. Box Num. | Open Available [250] | Supervised DL | EfficientDet | mAP50 = 90.1% | [251] |
| | | | | | Supervised DL | Modified YOLOv8 | mAP50 = 91.1% | [251] |

### 3) Substations and Transformers

Inspections of substations and transformers require reliable monitoring systems due to their critical role in T&D. Early detection of sudden temperature rises, which often precede equipment failures, is essential to prevent assets damage and power outages. Fixed platforms equipped with IRT and UV sensors enable real-time analysis of these assets, facilitating the immediate identification of anomalies such as overheating or insulation degradation. This proactive monitoring might reduces downtime and enhances operational efficiency.

Sheng Han *et al.* [252] developed a smart thermographic camera for diagnosing electrical equipment failures, based on three severity levels (emergency, serious, and minor), determined by thresholds of maximum temperature ($T_1$) and relative temperature difference (RTD), in accordance with the DL/T664 [262] standard, which provides application guidelines for infrared diagnosis of live electrical equipment. The authors used temperature data from 100 anomaly images to model a linear regression function relating RTD adjusted for environmental temperature to the RTD of the asset, achieving 92% accuracy compared to prior classification criteria. The dataset used comprises 12,683 infrared images from 37 substations, containing 18,146 annotated objects across 16 asset classes, including transformers, breakers, and arresters, at various voltage levels. Multiple models were trained, including a proposed simplification of YOLOv4, which achieved an mAP of 86.4% with an inference time of 33 ms. In comparison, RetinaNet achieved an mAP of 85.3% with an inference time of 45 ms on the validation set. The final step in their diagnostic system calculates the temperature difference between defective and functional assets. Compared to manual analysis using a FLIR T640 infrared camera, the smart camera correctly identified 92.2% of the 295 defective assets recorded.

Jianhua Ou *et al.* [253] used an improved Faster R-CNN model to detect substation electrical equipment in infrared images. They collected 1,263 images of five substation assets: current transformer (CT), voltage transformer (VT), arrester (AR), circuit breaker (CB), and insulator strings (IS). They reserved 245 images for testing and applied offline augmentations to the remainder for training and validation. An ablation study demonstrated improved performance for the three proposed modifications: using a VGG16 backbone, adding 3:1 and 1:3 anchor ratios, and discarding high-level convolution layers. The model achieved an mAP50 of 92.78% at 11 fps, while YOLOv4 achieved 91.85% at 13 fps. The model's robustness was also verified against Gaussian noise, salt-and-pepper noise, and brightness changes.

Bin Wang *et al.* [254] proposed an semi-automatic diagnosis method using instance segmentation and temperature analysis. They collected 1,200 IR images, and annotated polygonal insulator instances. The authors fine-tuned Mask R-CNN model with the ResNet101-FPN backbone, achieving a mAP50 of 77% on the test set. Their diagnostic system utilizes temperature data from the insulator mask pixels to compute maximum and average temperatures, determining faulty equipment based on explicit parametric conditions.

Recognizing the lack of a public dataset of IR images of insulators, Jinhui Zhou *et al.* [255] used a FLIR T640 IR camera to capture 2,540 images with $640 \times 480$ pixels, containing insulators on 110 kV transmission lines. They proposed a weakly supervised method to convert bounding box annotations into mask annotations, Box-Supervised Instance Segmentation (BSIS), utilizing a ResNet101-FPN backbone and a Shuffle-Polarized Self-Attention (SPSA) block. Their method achieved a box mAP of 95.19% and a mask mAP of 93.22%. While the dataset was not shared, limiting their current contribution to the scientific community, the authors plan to develop a multimodal dataset that incorporates temperature data from IR images, along with environmental temperature and wind data, for condition assessment in future work.

Besides disturbing the environment, wildlife, and nearby communications, the corona effect, caused by the ionization of the air, results in power loss and damage to assets. Prolonged exposure leads to wear and tear on insulating equipment through material degradation, ultimately causing energy inefficiencies and equipment failures.

Halil Ibrahim Uckol and Suat Ilhan [60] studied the visual light corona discharge patterns from different electrode tips under various voltage levels. They compared signals captured by a conventional digital camera to those from other corona sensors, including acoustic sensors, UV cameras, and electrical pulse visualization using a High-Frequency Current Transformer (HFCT). Their study describes the corona discharge phenomenon across several states and phases, including corona bursts, onset streamers, pulseless glow, and breakdown events. The authors identified several limitations in the sensors they tested. For example, since the HFCT cannot detect DC currents, it was also unable to identify positive glow discharges, which may occur during AC half-cycles. The acoustic signal, which is inaudible to humans, varies between phases and remains below the background noise level during the glow phase. The corona camera could easily detect streamer pulses but struggled with continuous glow, demonstrating difficulty in assessing the severity of corona discharges. Conversely, digital cameras were able to capture every stage of corona discharge through their corresponding light patterns. In their follow-up study [257], the authors advanced their research with a YOLOv8 detector and CNN ensemble classifier for DC corona discharge. They compiled the images from their experiments into a dataset shared on IEEE Dataport [256] and supplemented it with additional data from the literature. This resulted in a total of 741 images with annotated bounding boxes over four distinct corona discharge modes (A: -DC pulseless glow, B: Hermstein's glow, C: streamer, and D: Trichel pulse stages). The trained YOLOv8m model achieved a mAP50 of 99.3% in detecting corona purple flashes. The detected regions were then cropped and classified by an ensemble model combining the outputs of four CNNs (AlexNet, ResNet50, MobileNet, and ShuffleNet), achieving 100% accuracy on the unseen test set.

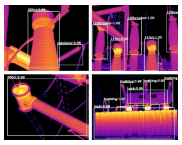
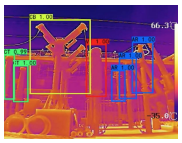
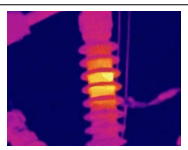
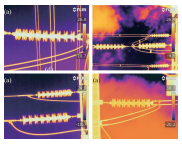
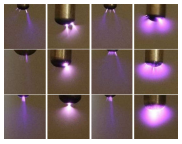
To overcome the limitations of detecting visible corona discharge during daylight, Noushin Davari *et al.* [258] pro-

posed an inspection system for identifying UV defects in MV power lines. The system begins by detecting insulator objects and uses their central X-coordinates to divide the image into three regions corresponding to the R, S, and T phases. By averaging the pulsing corona discharge pattern over 10 consecutive frames of the UV channel, the system reduced noise compared to their earlier work [51]. The authors augmented the original dataset of 462 images to 2,310 using artificial augmentation techniques. Despite concerns about dataset split integrity affecting the validation and test sets, the Iran DSO units reported satisfactory performance after deploying the software. The system achieved 94% accuracy, in 100 video records, against ground truth labels established by experts.

Recognizing the lack of annotated datasets for electrical substation equipment, Andreas Gomes *et al.* [261] created a diverse dataset using human, fixed, and AGV-mounted cameras at different times of the day. It contains 1,660 RGB images with resolutions from $640 \times 480$ to $4672 \times 3504$ pixels, captured at a single distribution substation in Brazil over two years and shared across several data portals [259], [263], [264]. The dataset includes 50,705 annotated polygons spanning 15 classes, such as insulators, transformers, breakers, and disconnect switches. It was designed to validate weakly supervised learning methods. The authors demonstrated usability by training a DeepLabV3+ model to predict segmentation masks from bounding box annotations. They employed a recursive refinement strategy using six iterations of different post-processing methods to improve the generated segmentation masks. Their best results, including an IoU of 62.9% for the "recloser" class, were achieved with a training set combining ground truth masks from the original dataset and segmentation masks generated in the fifth generation of the GrabCut SDI model. This result highlights the effectiveness of their approach in refining weakly supervised methods.

Table 14 summarizes the literature details and results on substations assets using IRT and UVI sensors on different platforms, with a focus on crucial details for DL model comparison, such as task definition and dataset characteristics.

**TABLE 14. Literature Details and Results on Substations and Transformers employing IRT Sensors on different Platforms**

| Task Description | Data | | | | Model | | | |
|---|---|---|---|---|---|---|---|---|
| | Sample | Size | Annotations | Source | Algorithm | Architecture | Performance | Ref |
| Substations Assets Health Condition Diagnostic |  | 12,683 Images 640 x 480 x 1 | 16 Classes 18,146 B. Box Num. | Not Available | Supervised DL | Retina-Net | mAP50 = 85.3% | [252] |
| | | | | | Supervised DL | Modified YOLOv4 | mAP50 = 86.4% | [252] |
| Substations Assets Detection |  | 1,263 Images 600 x 600 x 1 | 5 Classes Unspecified B. Box Num. | Not Available | Supervised DL | YOLOv5 CSPdarknet53 | mAP50 = 91.85% | [253] |
| | | | | | Supervised DL | Faster R-CNN VGG16 | mAP50 = 92.78% | [253] |
| HV Insulators Temperature Measurement |  | 1,200 Images 1024 x 1024 x 1 | 1 Classes Unspecified B. Box Num. | Not Available | Supervised DL | Mask R-CNN ResNet101-FPN | mAP50 = 77% | [254] |
| HV Insulators Segmentation |  | 2,540 Images 640 x 480 x 1 | 1 Classes Unspecified B. Box Num. | Not Available | Supervised DL | BSIS ResNet101-FPN | mAP50 = 95.19% | [255] |
| Corona Detection and Discharge Mode Classification |  | 741 Images 500 x 500 x 3 | 4 Classes Unspecified B. Box Num. | Open Available [256] | Supervised DL | YOLOv8m | mAP50 = 99.3% | [257] |
| | | | | | Supervised DL | ShuffleNet | F1 = 100% | [257] |
| Corona Detection and Power Equipment Classification |  | 462 Images 4 Channels Unspecified Resolution | 7 Classes 4,486 B. Box Num. | Not Available | Supervised DL | Faster R-CNN | mAP = 78% | [51] |
| | | | | | Supervised DL | Optimized Faster R-CNN | mAP = 82% | [258] |
| Substations Assets Instance Segmentation |  | 1,660 Images 640 x 480 x 3 4672 x 3504 x 3 | 15 Classes 50,705 B. Box Num. | Data + Code Available [259], [260] | Weak-Supervised DL | DeepLabV3+ GrabCut SDI | IoU = 62.9% | [261] |

## VI. CURRENT LIMITATIONS AND CHALLENGES

This review identifies key limitations in current practices that hinder the development of effective methodologies for automating electrical grid asset inspections. While common DL challenges emphasized in other studies—such as data annotation costs, computational efficiency, and model interpretability—are relevant across various domains, we argue that these are not the primary factors currently hindering the transition from human visual analysis to computer vision-based inspection of electrical grid assets.

Although well-recognized issues—such as data leakage and incomplete evaluation of models—-undermine the reliability of reported results, they remain common in recent literature. For example, applying augmentation before dataset splitting may introduce data leakage, as augmented variants can appear in both training and test sets. A more robust approach would involve applying augmentation after splitting or incorporating it into the training pipeline. Similarly, cascade architectures are often evaluated only at the component level, with limited reporting of end-to-end performance, making it difficult to assess overall system effectiveness.

Beyond these technical concerns, our broader interdisciplinary perspective highlights often-overlooked issues, such as the lack of methods for measuring data quality in model development and testing, which directly impacts the reliability of trained models. Additional challenges arise from a misalignment between academic research and industry needs, particularly in assessing the operational viability of automating high-value tasks. This disconnect extends to performance evaluation, where models are tested under conditions that do not adequately represent real-world scenarios, leading to unrealistic deployment expectations.

We argue that at its core, this misalignment stems from inadequate system decomposition. Without a structured approach to problem-solving, progress in automating electrical grid asset inspections remains limited. System decomposition is essential for managing complex problems, but if not done carefully, it can lead to critical gaps in requirements. When dividing a system into subsystems, some constraints may become implicit assumptions, while responsibilities for key aspects, such as data quality, can become unclear. In electrical grid asset inspections, this is particularly problematic if the different teams handling acquisition, processing, and model development do not effectively communicate their needs and constraints.

### A. DATA QUALITY MEASUREMENT

Classical examples of poor interface definition in system decomposition leading to suboptimal approaches include developing complex models to compensate for occlusions or glares, as well as attempting to detect small defects in low-resolution images rather than focusing on improving data acquisition strategies and ensuring higher-quality data collection. Failing to address the root causes of inspection challenges efficiently may result in unnecessary complexity in the system.

The effects of this lack of coordination in dataset construction lead to a shortage of methods for measuring data quality and representability, limiting efficient model development. A large dataset does not necessarily imply data diversity; it is crucial that the dataset adequately represents various defect types, lighting conditions, and viewing angles. Removing redundant or over-represented data can expedite the training process and mitigate performance degradation, while including informative data improves prediction accuracy and robustness [265]. Data quality also encompasses annotation quality, which is not only about minimizing noise through consistent annotations using thoughtful procedures and frameworks but also includes aspects like spatial granularity and semantic detail in category definitions [266].

To avoid these pitfalls, an integrated approach is essential. Rather than treating each stage in isolation, flexible interfaces and feedback loops should be incorporated to ensure that data acquisition, processing, and model training work cohesively to optimize inspection results. Prioritizing automation in high-risk scenarios and aligning dataset curation to reflect real-world defect distributions help ensure solutions are both technically sound and operationally valuable for grid operations.

### B. OPERATIONAL VIABILITY

While automated systems help reduce costs and improve consistency, they come with their own expenses and challenges, such as, development and maintenance costs, integration with existing systems efforts, or service failure handling overload [50]. Ensuring reliable scalability, and balancing false alarms with accurate defect detection remain ongoing concerns. While automation has the potential to surpass human performance, achieving this level of accuracy and adaptability is still a work in progress.

The decision to automate defect detection should consider both the frequency and impact of potential events. Automating the detection of rare, low-impact anomalies may be impractical, as the cost and complexity of developing such a system could outweigh the benefits. In these cases, periodic manual inspections or targeted sampling may be more efficient. Conversely, automating the detection of frequent, high-impact defects is highly valuable. Failures that disrupt system-wide operations or cause costly damages justify the investment in automated solutions, as early detection and rapid response can prevent cascading failures and reduce downtime. By strategically prioritizing automation for critical, high-risk events, organizations can maximize returns on investment while maintaining flexibility for less critical scenarios.

When an inspection system is poorly decomposed and data processing is optimized in isolation, operational viability is often overlooked. This misalignment between academic research and industry needs is can be seen in studies that rely on augmented [214], synthetic [207], [212], or simulated [208] data due to the lack of real-world examples. The absence of real data suggests low occurrence rates, raising doubts about the practical feasibility of such detection tasks.

Focusing solely on model development can escalate detection tasks to extreme cases, misaligning detection capabilities with actionable maintenance decisions. For example, detecting a monkey on an insulated power line [211] may even achieve high accuracy, but its operational relevance is questionable. Unlike bird nests, which are static and justify scheduled removal, a monkey on a PL is transient, and shutting down the line or dispatching a crew is impractical. Furthermore, handling wildlife raises legal and ethical concerns that fall outside the scope of grid maintenance. Optimizing data processing tasks alone is insufficient for developing automated systems. If detection triggers do not lead to meaningful maintenance actions, their operational value is questionable. Aligning automated detection with practical grid maintenance goals is crucial.

### C. PERFORMANCE EVALUATION

Many efforts focus on system verification, ensuring that the system works as intended, while insufficient attention is paid to system validation, which ensures that the system addresses the right problem. This misalignment often results in high-precision models that fail to bring value to stakeholders. For instance, optimizing bounding box regression accuracy might result in higher mAP, but in asset inspection, the defect's existence is far more important than its predicted bounding box score. Focusing solely on the technical correctness of the system, without validating its relevance to the problem it aims to solve, can lead to wasted resources and suboptimal field performance. Optimizing detection metrics can detract from a system's practical value. High benchmark scores do not guarantee that a model will be operationally useful. Furthermore, a model may perform well in terms of detection accuracy, but if it fails to translate detection into actionable maintenance responses, it holds little value for stakeholders.

The issue of overfitting is directly linked to the lack of proper system validation. DL models can overfit to specific datasets, particularly when trained on synthetic or augmented data. As a result, they struggle to generalize to new, unseen conditions in real-world scenarios. Without robust validation on diverse, representative data, system performance is likely to degrade when confronted with novel situations or edge cases not adequately represented in training. To mitigate these risks and enhance real-world applicability, a more effective strategy is to train models on smaller datasets while testing them on diverse and realistic data. This approach helps expose models to varied operational conditions during evaluation, providing a clearer picture of their generalization capability and reducing the likelihood of overfitting.

The lack of shared models, data, and code within the research and industrial communities hinders the ability to replicate and build upon previous work. This lack of standardization and transparency prevents the development of generalized solutions. Inconsistent results from different approaches further complicate the evaluation of model performance, and this fragmentation limits the potential for scaling up defect detection systems.

In practice, defects often exhibit class imbalance, with certain defect types occurring far more frequently than others. While data augmentation can improve performance across all classes, this may obscure the system's real-world value. Small gains in AP for frequent, high-impact defects are typically far more valuable than improvements in detecting rare, low-risk anomalies. To address this, performance metrics should reflect the operational needs of grid maintenance, prioritizing defects based on their frequency and impact. Aligning evaluation criteria with maintenance objectives ensures that detection leads to meaningful actions that optimize grid performance, reduce downtime, and enhance system reliability.

## VII. FUTURE DIRECTIONS

Automating electrical grid inspections with deep vision technologies holds transformative potential by improving reliability and safety while lowering operational costs. However, realizing this potential at scale requires addressing persistent challenges in data processing, model robustness, and system integration, enabling real-time condition monitoring, predictive maintenance, and a more resilient infrastructure.

To address the challenges and limitations in current practices that hinder the development of effective methodologies for automating electrical grid asset inspections, we propose in this work, an innovative data-centric approach, illustrated in Fig. 16, that integrates embedded vision for on-site dataset validation, multimodal learning for domain knowledge transfer, and active learning strategies for efficient model adaptation.

### A. EMBEDDED VISION

Popular publicly available datasets are suitable for model fitting experiments but are limited for real system development. To overcome common issues in data capture, such as blurring and glare, we propose an embedded vision system to validate the quality and quantity of acquired images, ensuring full coverage of the asset surface.

The yellow diagram blocks in Fig. 16 illustrate the following process. Given an ***Inspection Order*** (green block) specifying the asset to be inspected and the means employed (as previously described in Fig. 11), the platform operator performs the ***Flight Control*** to an optimal position for ***Image Acquisition***. The image ***Quality*** is validated by an embedded vision system, evaluating attributes such as sharpness and contrast, preventing the collection of ***New Data*** affected by excessive brightness or poor visibility. The operator continues capturing valid images while the embedded vision system compute the asset's surface coverage, ensuring that a sufficient ***Quantity*** of data is collected before finalizing the on-site ***Data Collection***.

Since experienced operators can often perform these validation tasks effectively, automation at this stage may be challenging to justify, as it does not provide immediate value. However, integrating such capabilities into autonomous flying vehicles is essential for enabling the next generation of asset inspection systems.

### B. MULTIMODAL DATA

As vision models evolve into multimodal large-scale architectures, their strong performance in general-purpose tasks is largely attributed to training on diverse Internet-sourced datasets. However, their effectiveness in specialized contexts, such as electrical grid asset defect detection, remains limited due to a lack of domain-specific knowledge and tailored training data [267].

To bridge this gap, we propose integrating multiple data modalities into a comprehensive dataset, maximizing the potential of multimodal large language models (MLLMs).

The orange diagram blocks in Fig. 16 illustrate the following process. For instance, ***Time-Series Data***, such as local ***Climatic Data*** and ***Electrical Measurements***, can complement infrared ***Thermographic Data***, enhancing thermal analysis accuracy. Similarly, structured system data, including power line topology, voltage levels, number of circuits, and insulator types, can guide classification models by constraining predictions to valid asset categories.

Different visual modalities can also enhance one another. ***3D Point Clouds*** and depth maps improve segmentation by distinguishing foreground assets from the background, thereby aiding object detection in ***Visual Data***. Additionally, ***Maintenance Records*** can support the development of a domain-specific linguistic dictionary, while their statistical patterns may help balance datasets, ensuring a more representative learning process.

The inclusion of domain-specific multimodal data facilitates knowledge transfer, improving model adaptability to real-world inspection scenarios.

### C. ACTIVE LEARNING

The shift from large datasets to high-quality data is particularly relevant in industries with limited data availability [135]. Additionally, labeling errors in popular benchmark datasets [136] have highlighted new opportunities in data-centric AI [37]. Active learning methodologies and tools [137], [138] can reduce the need for labeled data by prioritizing the most informative examples.

The red diagram blocks in Fig. 16 illustrate our proposed implementation of an active learning methodology applied to an automated electrical grid asset inspection system. ***Unlabeled Data*** are processed by the latest ***Deployed Model***, which performs ***Inference*** to generate an ***Output*** containing predicted labels. These predictions are ranked by confidence and subsequently validated by a ***Human*** oracle, optimizing the annotation pipeline. The ***Labeled Data*** is then subsampled by a ***Selection*** function, which computes feature similarity within the latent space to prioritize samples that maximize representation across the function space. This ensures a diverse ***Training Dataset***, leading to more effective gradient updates, improved generalization, and minimized loss during ***Model Training***. This iterative ***Active Learning Loop*** enables continuous model refinement, integrating newly labeled data into successive training cycles to produce incrementally improved ***Deployed Model*** versions.

Finally, the purple diagram blocks in Fig. 16 illustrate the reporting process. ***Dynamic*** and ***Periodic*** updates of all collected data, including inspection findings and real-time ***Electrical Measurements***, can be integrated with ***Geospatial Data*** into a ***Digital Twin*** interface, enabling more effective and interactive asset condition assessment.

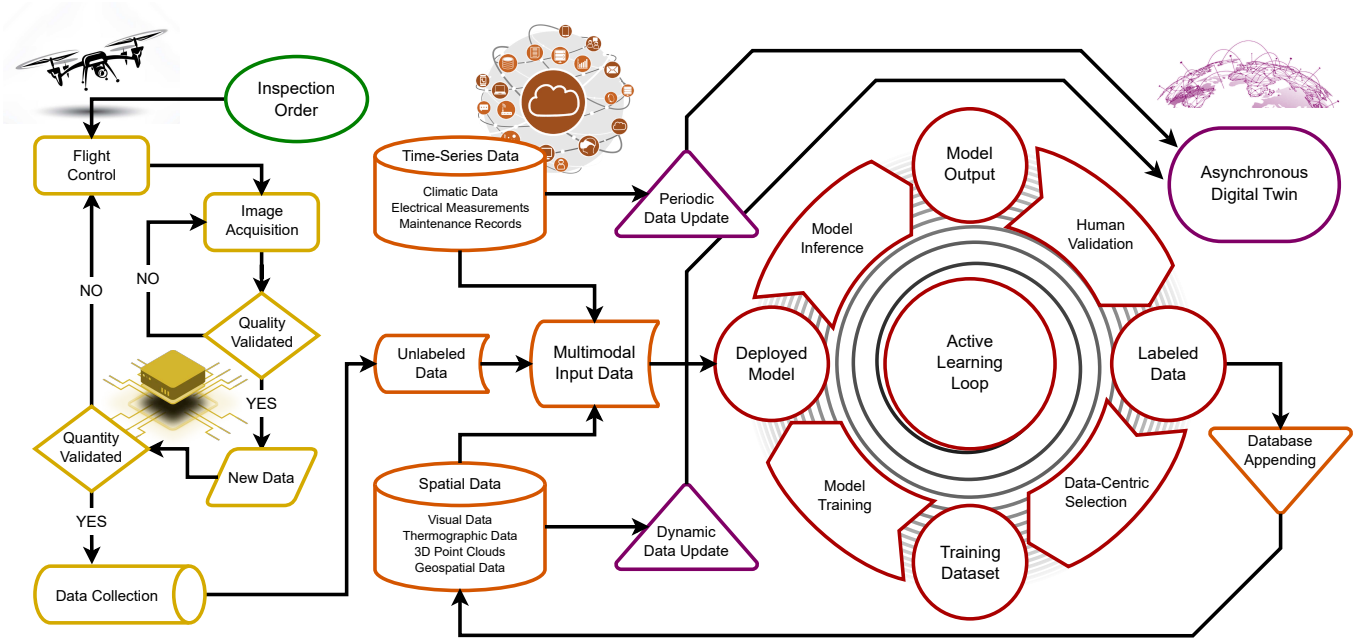



**FIGURE 16.** Proposed automated ecosystem integrating embedded vision and data-centric multimodal active learning methodologies. The embedded vision system validates both the quantity and quality of the acquired data. Spatial and time-series data are combined into a multimodal input to leverage domain knowledge transfer. The model's iterative development incorporates a human validator and training data selection within an active learning paradigm. Efficient reporting of asset conditions is achieved through a digital twin interface.

## VIII. CONCLUSION

This review explored the evolving landscape of electrical grid asset inspection, offering a comprehensive synthesis of current technologies, application domains, and research trends.

The discussion began by analysing the economic and political motivations driving the development of efficient inspection systems. This was followed by a foundational overview of electrical grid systems typically targeted in automated visual inspections, along with a characterization of the inspected assets, highlighting how inspection strategies must adapt to asset-specific constraints and other operational factors.

Throughout the document, the main components of an automated inspection system are examined, discussing key concepts, benefits, and limitations, and highlighting their roles and interactions within the overall inspection pipeline. The review compares the characteristics of various data acquisition technologies and platforms, and summarizes the development of deep learning models, framed by their historical evolution and emerging paradigms in deep vision.

Literature was reviewed across application-specific scenarios, ranging from renewable power generation to transmission and distribution systems. These analyses highlighted shared practices and persistent challenges across use cases, such as distinguishing true generalization from overfitting, due to limited dataset diversity and inconsistent validation. These issues are often rooted in a misalignment between goals of model-centric approaches and practical requirements.

Current performance evaluation practices tend to overemphasize benchmark scores while undervaluing operational metrics that more accurately reflect the impact of defects on grid maintenance. This tendency overlooks the need for inspection systems that prioritize actionable outcomes over test accuracy alone, ensuring that defect detection directly supports meaningful maintenance decisions.

The study outlines promising future directions, emphasizing effective integration of emerging technologies to address the identified challenges. These directions include embedded vision systems for promptly validating data at the point of capture, multimodal data fusion to leverage large language model architectures, and active learning strategies to reduce annotation burdens, support efficient training, and enhance model robustness. Together, these technologies represent key steps toward scalable, adaptive, and operationally viable inspection solutions for electrical grid assets.

Looking ahead, bridging the gap between academic research and industrial application is essential. Future efforts should prioritize shared models, open datasets, and closer collaboration between researchers and utility operators. Only through such alignment can the field progress toward automated inspection systems that are not only accurate but also capable of supporting informed maintenance decisions.

Ultimately, this review aims to guide future research toward impactful and deployable innovations in automated inspection systems, ensuring that emerging technologies support the sustainable operation of electrical infrastructure while addressing the practical needs of grid asset management.

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